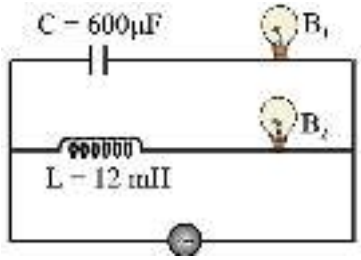
PHYSICS

1) In the circuit shown in figure, if both the bulbs B_1 and B_2 are identical then :-



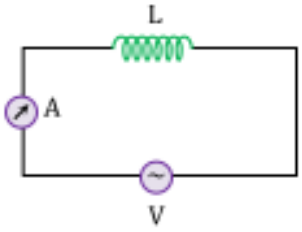
(1) Both bulbs have equal brightness.

(2) B_2 will be brighter than B_1.

(3) As the frequency is increased, brightness of B_1 will increase and that of B_2 will decrease.

(4) As the frequency is decreased the brightness of B_1 will increase and that of B_2 will decrease.

2) In given circuit applied voltage is $V = 50\sqrt{2}$ sin (100πt) volt and ammeter reading is 2A then



calculate value of L?

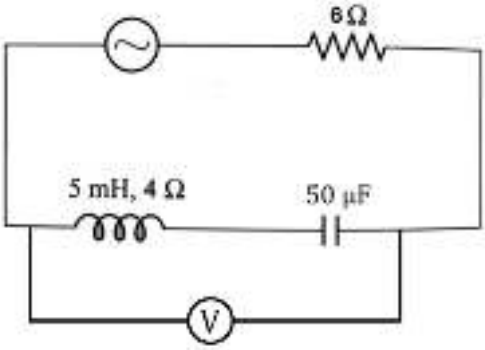
(1) $\frac{1}{4\pi}$H

(2) $\frac{1}{2\pi}$H

(3) $\frac{1}{\pi}$H

(4) 2πH

3) In the circuit shown below, the ac source has voltage V = 20 cos(ωt) volts with ω = 2000 rad/sec.



The reading of AC voltmeter.

(1) 5.6V

(2) 8V

(3) Zero

(4) 20V

4) If V = 200sin $\left[\omega t + \frac{\pi}{6}\right]$ volt and $I = 10 \sin\left[\omega t + \frac{\pi}{3}\right]$ ampere. Calculate average power consumed in AC circuit.

(1) $500\sqrt{2}W$

(2) $500\sqrt{3}W$

(3) $200\sqrt{3}W$

(4) $300\sqrt{3}W$

5) Match Columns I and II for a.c. circuits.

	Column I		**Column II**
(A)	RL circuit	(1)	Leading quantity - current
(B)	RC circuit	(2)	Leading quantity - voltage
(C)	Pure Inductive circuit	(3)	Phase difference between voltage and current 0°
(D)	Pure Resistive circuit	(4)	Phase difference between voltage and current 90°

(1) (A)→(2) ; (B)→(1) ; (C)→(3) ; (D)→(4)

(2) (A)→(2) ; (B)→(2) ; (C)→(4) ; (D)→(3)

(3) (A)→(4) ; (B)→(3) ; (C)→(2) ; (D)→(1)

(4) (A)→(2) ; (B)→(1) ; (C)→(4) ; (D)→(3)

6) If $i = t^2$, for $0 < t < T$ r.m.s. value of current is:

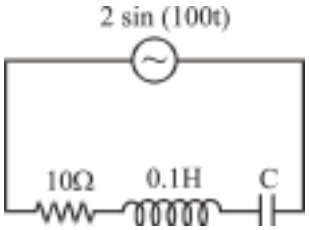
(1) $\frac{T^2}{\sqrt{2}}$

(2) $\frac{T^2}{2}$

(3) $\frac{T^2}{\sqrt{5}}$

(4) None of these

7) The power factor of circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to :-



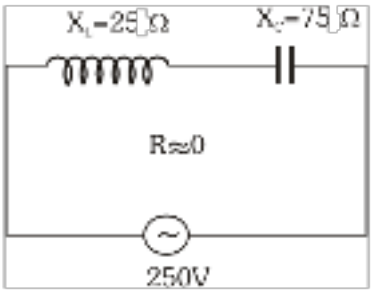
(1) 400 µF

(2) 300 µF

(3) 500 µF

(4) 200 µF

8) For shown circuit :-



(a) Current in circuit is 10 A
(b) Voltage across inductor is 125 V
(c) Voltage across capacitor is 375 V
(d) Current in circuit in 5A.
Choose the correct option :

(1) a,b,c

(2) a,c

(3) b,c,d

(4) a,b

9) The self inductance of a choke coil is 10 mH. When it is connected with a 10V dc source, then the loss of power is 20 watt. When it is connected with 10 volt ac source, then loss of power is 10 watt. The frequency of ac source will be approximately.

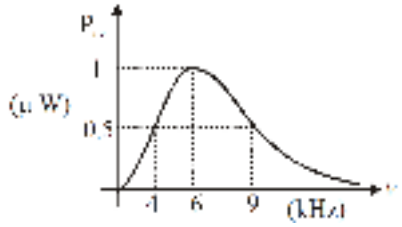
(1) 50 Hz

(2) 60 Hz

(3) 80 Hz

(4) 100 Hz

10) The plot given below is of the average power delivered to an LRC circuit versus frequency. The quality factor of the circuit is :



(1) 5

(2) 2.4

(3) 2.8

(4) 1.2

11) Which of the following have zero average value in a electromagnetic wave? (a) electric field
(b) magnetic field
(c) electric energy
(d) magnetic energy

(1) a, d

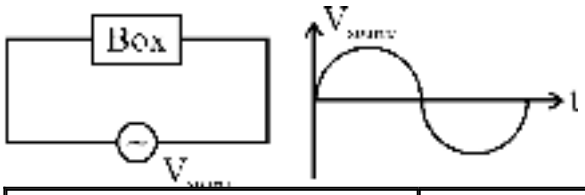
(2) c, d

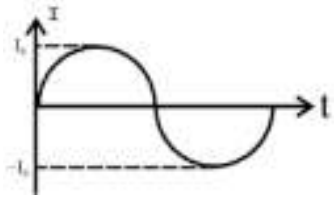
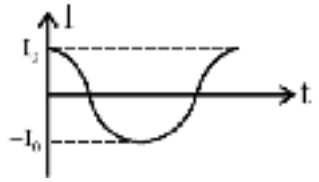
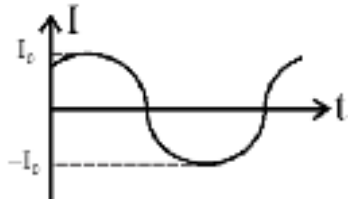
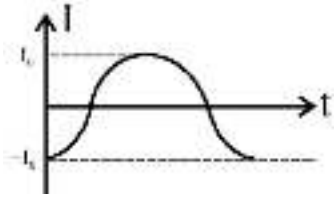
(3) a, b

(4) a, c

12)

Box may have any series combination of L, C and R.



Column-I		Column-II	
A		P	Box may contain LCR
B		Q	Box may contain only LR
C		R	Power factor of box is zero
D		S	Box may contain only CR
		T	State of resonance

(1) A →P, T; B→ R; C → P, S; D → R

(2) A → P, R; B → S; C → T; D → P, Q

(3) A → P,S; B →R; C → T; D →P, Q

(4) A → P, T; B → T; C → R; D → P, Q

13) In an LCR circuit R = 100 ohm. When capacitance C is removed, the current lags behind the voltage by $\frac{\pi}{4}$. When only inductance L is removed out of these the current leads the voltage by $\frac{\pi}{4}$. The impedance of the circuit is

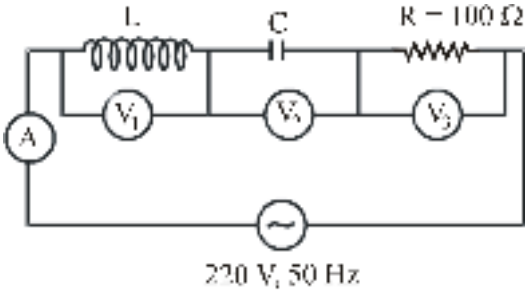
(1) 50 ohm

(2) 100 ohm

(3) 200 ohm

(4) 400 ohm

14) In the given circuit the reading of voltmeter V_1 and V_2 are 300 volts each. The reading of the voltmeter V_3 and ammeter A are respectively :



(1) 100 V, 2.0 A

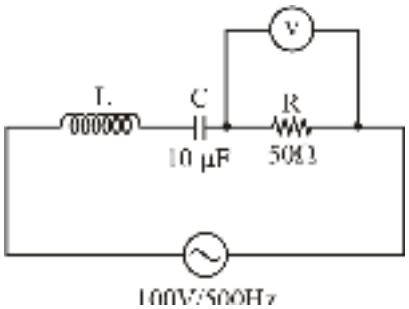
(2) 150 V, 2.2 A

(3) 220 V, 2.2 A

(4) 220 V, 2.0 A

15)

If in the circuit shown, voltmeter reads 100 V then L is :-



(1) 0.1 Henry

(2) 0.2 Henry

(3) 0.02 Henry

(4) 0.01 Henry

16) In an LCR circuit, capacitance is changed from C to 2C. For the resonant frequency to remain unchanged, the inductance should be changed from L to –

(1) 4 L

(2) 2 L

(3) L/2

(4) L/4

17) An AC having a peak value 1.41A is used to heat a wire. For producing same heat value of DC will be-

(1) 1.41 A

(2) 2.0 A

(3) 0.707 A

(4) 1.0 A

18) A plane electromagnetic wave propagating in the x-direction has wavelength of 6.0 mm. The electric field is in the y-direction and its maximum magnitude is 33 V m^{-1}. The equation for the electric field as a function of x and t is

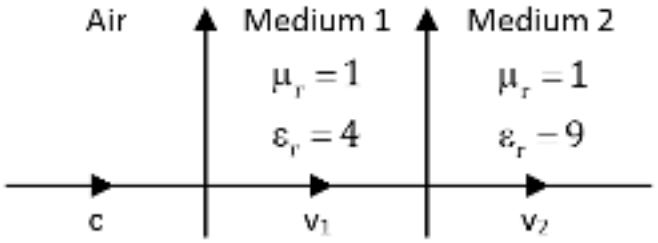
(1) $11\sin\left[\pi\left(t-\frac{x}{c}\right)\right]$

(2) $33\sin\left[\pi\times10^{11}\left(t-\frac{x}{c}\right)\right]$

(3) $33\sin\left[\pi\left(t-\frac{x}{c}\right)\right]$

(4) $11\sin\left[\pi\times10^{11}\left(t-\frac{x}{c}\right)\right]$

19) As shown in the figure, after passing through the medium 1, the speed of light v_2 in medium 2 will be
(Given c = 3 $\times10^8$ ms^{-1})



(1) 1.0 $\times10^8$ ms^{-1}

(2) 0.5 $\times10^8$ ms^{-1}

(3) 1.5 $\times10^8$ ms^{-1}

(4) 3.0 $\times10^8$ ms^{-1}

20) Match List-I with List-II

	List-I (Electromagnetic waves)		**List-II (Wavelength)**
(a)	Gamma rays	(i)	10^{-5} m

(b)	UV rays	(ii)	10^{-12} m
(c)	Infrared radiations	(iii)	10^{3} m
(d)	Radio waves	(iv)	10^{-8} m

Choose the correct answer from the options given below:

(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(2) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(3) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

21) An electromagnetic wave going through vacuum is described by $E = E_0 \sin(kx - \omega t)$. Which of the following is/are independent of the wavelength?

(1) k

(2) ω

(3) k/ω

(4) kω

22) Two persons of equal height are carrying a long uniform wooden beam of length □. They are at distance □/4 and □/6 from nearest ends of the rod. The ratio of normal reactions at their heads is n : 3 where n is :

(1) 4

(2) 6

(3) 8

(4) 9

23) Moment of inertia of a body about a given axis is 1.5 kg m^2. Initially the body is at rest. In order to produce a rotational kinetic energy of 1200 J, the angular acceleration of 20 rad/s^2 must be applied about the axis for a duration of :-

(1) 2 s

(2) 5 s

(3) 2.5 s

(4) 3 s

24) **Assertion:** A ladder is more apt to slip, when you are high up on it than when you just begin to climb.
Reason: At the high up on a ladder, the torque is large and it is small when one just begins to climb.

(1) Both assertion and reason are true and the reason is the correct explanation of the assertion.

(2) Both assertion and reason or true but reason is not the correct explanation of the assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

25) A constant torque acting on a uniform circular wheel changes its angular momentum from A_0 to $4A_0$ in 4 sec. the magnitude of this torque is :

(1) $4A_0$

(2) A_0

(3) $3A_0/4$

(4) $12A_0$

26) A disc of mass M and radius R is reshaped in the form of ring of same mass but radius 2R. The radius of gyration increases by a factor :

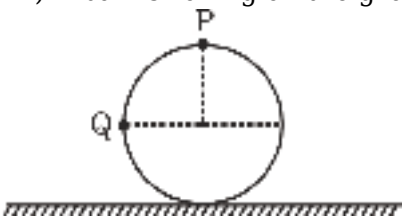
(1) $\sqrt{2}$

(2) $2\sqrt{2}$

(3) 4

(4) 2

27) A ball is rolling on the ground. The ratio of speed at points P and Q is :



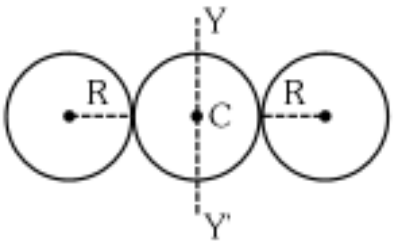
(1) $\sqrt{2}:1$

(2) 2 : 1

(3) $2\sqrt{2}:1$

(4) 1 : 2

28) Three spherical shells each of mass M and radius R are kept as shown. Find the moment of



inertia of the system about axis YY'.

(1) 8 MR^2

(2) 12 MR^2

(3) 14 MR^2

(4) 10 MR^2

29) A thin rod of mass m and length is oscillating about an axis passing through its one end. Its maximum angular speed is ω. Its centre of mass will rise upto maximum height :-

(1) $\frac{1}{6}\frac{\ell\omega}{g}$

(2) $\frac{1}{2}\frac{\ell^2\omega^2}{g}$

(3) $\frac{1}{6}\frac{\ell^2\omega^2}{g}$

(4) $\frac{1}{3}\frac{\ell^2\omega^2}{g}$

30) A wheel starting from rest is uniformly accelerated at 2 rad/s^2 for 20 seconds. It is allowed to rotate uniformly for the next 10 seconds and is finally brought to rest in next 20 seconds. The total angle rotated by the wheel (in radian) is :-

(1) 600

(2) 1200

(3) 1800

(4) 300

31) A disc is rotating with angular velocity $\vec{\omega}$. A force $\vec{F}$ acts at a point whose position vector with respect to the axis of rotation is $\vec{r}$. The power associated with torque due to the force is given by :

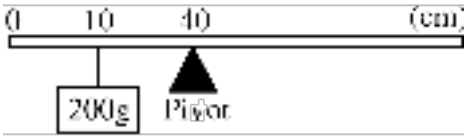
(1) $(\vec{r} \times \vec{F}) \cdot \vec{\omega}$

(2) $(\vec{r} \times \vec{F}) \times \vec{\omega}$

(3) $\vec{r} \times (\vec{F} \cdot \vec{\omega})$

(4) $\vec{r} \cdot (\vec{F} \times \vec{\omega})$

32) A uniformly graduated meter scale of mass 400 g with centre of gravity at 50.5 cm mark is pivoted at 40 cm. A 200 g mass is suspended at 10 cm mark. The position, where a 50g mass must be suspended so as to balance the scale is :



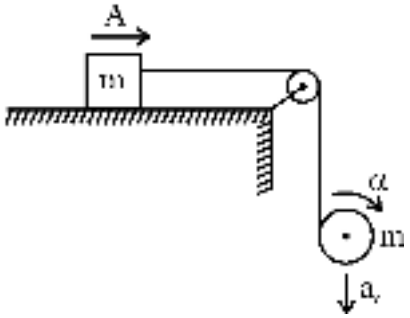
(1) 56 cm

(2) 64 cm

(3) 76 cm

(4) 80 cm

33) In the shown figure a cube of mass m is attached with a cylinder of same mass and radius r through an ideal string-pulley system. The string is wound over the cylinder. If α is the angular acceleration of the cylinder then the linear acceleration of cylinder a_c and the linear acceleration of cube A, are related as :



(1) $a_c = A$

(2) $A = a_c/2$

(3) $a_c + r\alpha = A$

(4) $a_c = A + r\alpha$

34) A rod of length L and mass M is bent to form a semicircular ring as shown in the adjoining figure. The moment of inertia about XY is :-



(1) $\frac{ML^2}{2\pi^2}$

(2) $\frac{ML^2}{\pi^2}$

(3) $\frac{ML^2}{4\pi^2}$

(4) $\frac{2ML^2}{\pi^2}$

35) To maintain a rotor at a uniform angular speed of 200 rad s^{-1}, an engine needs to transmit a torque of 180 N-m. What is the power required by the engine ?

(1) 12 kW

(2) 72 kW

(3) 36 kW

(4) 48 kW

36) A spherical shell rolls on a rough inclined plane in first case and slips down without rolling on smooth inclined plane of same length and same inclination in second case. The ratio of its accelerations in two cases will be :-

(1) $\frac{5}{3}$

(2) $\frac{3}{5}$

(3) $\frac{15}{13}$

(4) $\frac{13}{15}$

37) When torque is acting on a body, then body :-

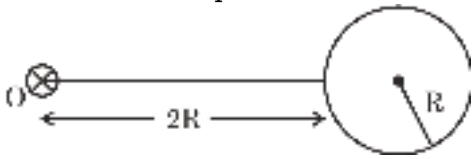
(1) Continue in the state of rest or uniform motion along a straight line

(2) Gets linear acceleration

(3) Rotates at constant speed

(4) Gets angular acceleration

38) A rod of mass m and length 2R is rigidly joined with a ring of mass m and radius R. This system is released in vertical plane from the horizontal position as shown and it is free to rotate smoothly about the fixed point 'O'. The initial angular acceleration of the system will be :



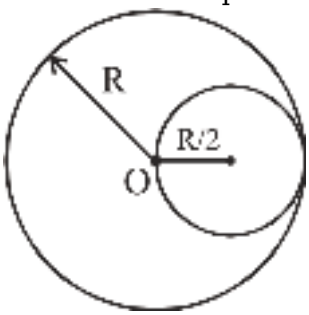
(1) $\frac{6g}{17R}$

(2) $\frac{3g}{2R}$

(3) $\frac{6g}{13R}$

(4) $\frac{2g}{3R}$

39) A circular hole of radius $\frac{R}{2}$ is cut from a circular disc of radius R and mass M according to the figure. Find moment of inertia of the remaining part about an axis passing through the centre of



complete disc and perpendicular the plane of disc

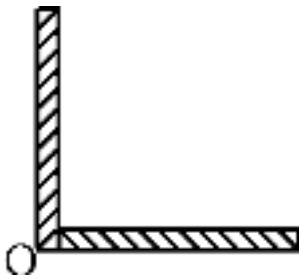
(1) $\frac{13}{32}MR^2$

(2) $\frac{26}{32}MR^2$

(3) $\frac{1}{32}MR^2$

(4) $\frac{1}{4}MR^2$

40) Two identical rods each of mass M and length L are kept according to figure. The moment of inertia of rods about an axis passing through O and perpendicular to the plane of rods, is :-



(1) $\frac{1}{3}ML^2$

(2) $\frac{2}{3}ML^2$

(3) $2ML^2$

(4) $\frac{1}{2}ML^2$

41) ***Assertion :***- A body rolls down an inclined plane without slipping. The fraction of total energy associated with its rotation will depend on its radius of gyration.
Reason :- Total kinetic energy of rolling body is equal to addition of kinetic energy of rotation and kinetic energy of translation.

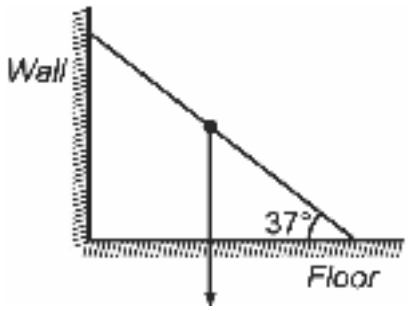
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

42) A uniform ladder of length 5 m and mass 200 kg is in equilibrium between vertical smooth wall and rough horizontal surface. Find minimum friction co-efficient between floor and ladder for this equilibrium.



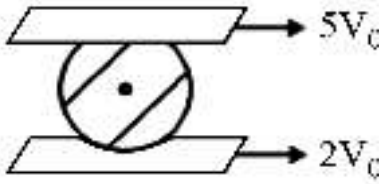
(1) 1/2

(2) 3/4

(3) 1/3

(4) 2/3

43) A cylinder of radius R is in state of rolling under shown condition, find angular velocity of cylinder :



(1) $\frac{3v_0}{2R}$

(2) $\frac{2v_0}{3R}$

(3) $\frac{3v_0}{4R}$

(4) $\frac{v_0}{2R}$

44) The rotational kinetic energy of two bodies of moments of inertia 9 kg-m^2 and 1kg-m^2 are same. The ratio of their angular momentum is :-

(1) 3 : 1

(2) 1 : 3

(3) 9 : 1

(4) 1 : 9

45) John is standing with folded hands at the centre of a platform rotating about its central axis. The kinetic energy of the system is K. Now, John stretches his arms so that moment of inertia of the system double. Kinetic energy of the system now is :-

(1) 2K

(2) 4K

(3) K/4

(4) K/2

CHEMISTRY

1) In inert pair effect, which pair of electrons are said to be inert:

(1) 2 electrons of (n–1)s

(2) 2 electrons of (n–1)d

(3) 2 electrons of np

(4) 2 electrons of ns

2) The ion that cannot undergo disproportionation is

(1) ClO_4^-

(2) ClO_3^-

(3) ClO_2^-

(4) ClO^-

3) Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.
Assertion (A) : The bond angle of NH_3 is greater than BiH_3.
Reason (R) : 'Bi' is metal while 'N' is non-metal.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are correct and (R) is the correct explanation of A.

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is incorrect

(4) Both (A) and (R) are incorrect.

4) Sulphur dioxide may be recognized by its?

(1) Characteristic pungent smell of burning sulphur
(2) Ability to turn dichromate paper green
(3) Ability to decolourize acidified $KMnO_4$ solution
(4) All

5) Which of the following is not correctly matched?

(1) Oxides of boron (B_2O_3) and silicon (SiO_2) are acidic in nature.
(2) oxides of aluminium (Al_2O_3) and gallium (Ga_2O_3) are amphoteric in nature.
(3) Oxides of indium (In_2O_3) and thallium (Tl_2O_3) are basic in nature.
(4) Oxides of germanium (GeO_2) and tin (SnO_2) are acidic in nature.

6) The IUPAC name of complex $K_3[Al(C_2O_4)_3]$ is;

(1) potassium aluminoxalate
(2) potassium tris(oxalato) aluminate(II)
(3) potassium aluminium (III) oxalate
(4) potassium trisoxalatoaluminate (III)

7) Of the following, which ligand does not posses the name suggested by IUPAC when it acts as a ligand in complex?

(1) H_2O, aqua
(2) NH_3, ammonia
(3) CO, carbonyl
(4) F^-, fluoro

8) The compound which does not show paramagnetism is;

(1) $[Cu(NH_3)_4]Cl_2$
(2) $[Ag(NH_3)_2]Cl$
(3) NO
(4) NO_2

9) What are the geometric shape and the oxidation number of the copper atom, respectively, for the complex ion,
$[Cu(NH_3)_4(H_2O)_2]^{+2}$?

(1) Tetrahedral; +2
(2) Square planar; –2
(3) Linear; +3

(4) Octahedral; +2

10) The solution of the complex $[Cu(NH_3)_4]SO_4$ in water:

(1) Will give the tests of Cu^{2+} ions.

(2) Will give the tests of NH_3.

(3) Will give the tests of SO_4^{2-} ions.

(4) Will not give the tests of any of the above species.

11) How many of following metals impart a characteristic colour to the Bunsen flame?
K, Ba, Sr, Mg, Be, Ca, Cu

(1) 1

(2) 6

(3) 7

(4) 5

12) Salt + $AgNO_3$ → (A)
Yellow ppt.
A is almost insoluble in concentrated ammonia solution. identify the anion in the salt.

(1) Cl^-

(2) S^{-2}

(3) I^-

(4) CH_3COO^-

13) Experiment of detection of S^{-2} is given below in alkaline medium
$S^{2-} + [Fe(CN)_5NO]^{2-} \rightarrow [Fe(CN)_5NOS]^{4-}$
Select the correct statement about the experiment.

(1) Colour of the complex compound formed in product is purple violet

(2) Oxidation number of Fe in complex compound formed in product is +3

(3) IUPAC name of complex compound formed in product is pentacyanothionitrito ferrate(I) ion

(4) Colour of complex formed is yellow

14) Which one of the following reactions does not occur ?

(1) $F_2 + 2Cl^- \rightarrow 2F^- + Cl_2$

(2) $Cl_2 + 2F^- \rightarrow 2Cl^- + F_2$

(3) $Br_2 + 2I^- \rightarrow 2Br^- + l_2$

(4) $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$

15) Which of the following is not peroxide ?

(1) Na_2O_2

(2) CaO_2

(3) PbO_2

(4) H_2O_2

16) There is no S—S linkage in

(1) $S_2O_4^{2-}$

(2) $S_2O_5^{2-}$

(3) $S_2O_3^{2-}$

(4) $S_2O_7^{2-}$

17) Which of the following statements are correct for complex $K_2[Zn(CN)_2(O_2)_2]$?
(I) IUPAC name : Potassium bisuperoxidobicyanidozinc (II).
(II) Number of geometrical isomers = 2.
(III) It is an anionic complex
(IV) It shows linkage isomerism.
(V) It is inner orbital complex.
Choose the correct option form given option:

(1) III and IV only

(2) I, II, III and IV only

(3) I, II, III, IV and V

(4) I, II and IV only

18) Which of the following statements is/are correct.
(i) In 3d-series, all metals form ionic oxides "MO' except Sc.
(ii) Basic strength order of oxides of vanadium are $V_2O_3 > V_2O_4 > V_2O_5$.
(iii) CrO is basic while Cr_2O_3 is acidic in nature.
(iv) Mn (VI) shows disproportionation to Mn (VII) & Mn (IV) in acidic medium.

(1) (i) only

(2) (ii), (iii)

(3) (i), (ii)

(4) (i), (ii), (iv)

19) Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.
Assertion (A) : Geometrical isomerism is not possible in tetrahedral complex having two different unidentate ligands coordinated with central metal ion.
Reason (R) : Tetrahedral complexes do not show geometrical isomerism because the relative position of unidentate ligands attach to central metal atom are same w.r.t. each other.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true, R is false.

(4) A is false, R is true.

20) x = Number of correct properties of d-block elements (3d series only).
y = Number of incorrect properties of d-block elements (3d series only).
Properties of d-block elements are given below:-
(1) V have max $\Delta H_{atomisation}$
(2) Ti have min $\Delta H_{ionisation}$ (I)
(3) Sc have max metallic radii
(4) E° value for M^{+2}/M is negative for V and Cr
(5) Density of Cu ≃ density of Ni
Find out x – y + 1 = ?

(1) 3

(2) 2

(3) 4

(4) 1

21)

Lanthanide contraction means:

(1) Contraction of atoms of elements after lanthanum due to poor shielding of the f-subshell electron.

(2) Contraction of atoms of elements before lanthanum due to poor shielding of the f-subshell electron.

(3) Contraction of an atom of lanthanum element due to poor shielding d-subshell electron.

(4) Contraction of an atom of lanthanum element due to high shielding of the d-subshell electron.

22) The geometry of $XeOF_4$ by VSEPR theory is:

(1) Square pyramidal

(2) Pentagonal planar

(3) Trigonal bipyramidal

(4) Octahedral

23) Given below are two statements , one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion :- Compounds in '+4' oxidation state of carbon are generally covalent in nature.
Reason :- Sum of first four I.P. is very high.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

24) In which of the following reactions the hybridisation state of underlined atom is same in reactants and products?

(1) $\underline{N}H_3 + H^+ \rightarrow NH_4^+$

(2) $\underline{Al}Cl_3 + Cl^- \rightarrow AlCl_4^-$

(3) $\underline{N}O_3^- + Zn + OH^- \rightarrow NH_3 + ZnO_2^{2-} + H_2O$

(4) $\underline{B}F_3 + F^- \rightarrow BF_4^-$

25) Which of the following is a bidentate ligand?

(1) en

(2) $EDTA^{4-}$

(3) dien

(4) trien

26) Which of the following is correct for the complex $[NiBr_2(PPh_3)_2]$?

(1) It is tetrahedral with two unpaired electrons.

(2) It is tetrahedral with one unpaired electron.

(3) It is square planar and diamagnetic.

(4) It is square planar with one unpaired electron.

27) Bucky ball or Buckminster fullerene is:

(1) it has sp^2-hybridised nature and resembles with a soccer ball

(2) an allotrope of carbon

(3) it is referred as C-60

(4) All of these

28) In the coordination compound $K_4[Ni(CN)_4]$, the oxidation state of nickel is:

(1) -1

(2) 0

(3) +2

(4) +1

29) The following statements are TRUE about crystal field theory except:

(1) It does not explain π bonding in complexes.

(2) The satisfactory explanation is not provided for the fact that water is a stronger ligand than OH^-.

(3) Partial covalent nature of metal ligand bond is not explained.

(4) It considers s, p and d orbitals of the central metal.

30) In SiF_6^{2-} and $SiCl_6^{2-}$, which one is known and why?

(1) SiF_6^{2-} because of small size of F

(2) SiF_6^{2-} because of large size of F

(3) $SiCl_6^{2-}$ because of small size of Cl

(4) $SiCl_6^{2-}$ because of large size of Cl

31) Amongst the following the most stable compound is:

(1) $[Fe(H_2O)_6]^{3+}$

(2) $[Fe(C_2O_4)_3]^{3-}$

(3) $[Fe(NH_3)_6]^{3+}$

(4) $[FeCl_6]^{3-}$

32) Which of the following is an organometallic compound:

(1) cis-platin

(2) Zeise salt

(3) Tollen's reagent

(4) Brown ring complex

33) For the confirmatory test of bromide ion, we acidify 1 mL of sodium carbonate extract with dil. HNO_3 (or take 1 mL water extract) and add silver nitrate solution. A pale yellow precipitate is obtained of-

(1) HBr

(2) AgBr

(3) AgI

(4) AgCl

34) Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.
Assertion (A) : Mohr's salt can be prepared using dilute HCl instead of dilute H_2SO_4.
Reason (R) : Chloride ions do not affect Fe^{2+} ions.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation

(3) A is true but R is false

(4) both A and R are false

35) Find the incorrect match:

(1) Al_2Cl_6 : 3C-4e bond is present

(2) $Al_2(CH_3)_6$: All carbon atoms are sp^3-hybridized

(3) I_2Cl_6 : Nonplanar

(4) Al_2Br_6 : Nonpolar

36) Among Fe^{3+}, Pb^{2+}, Cu^{2+} and Ni^{2+} identify the one that gets precipitated out while passing H_2S in presence of NH_4OH as group reagent. The highest possible oxidation state of the corresponding metal is

(1) +3

(2) +4

(3) +2

(4) +7

37) Which of the following complex formed by Cu^{+2} ions is most stable?

(1) $Cu^{+2} + 4NH_3 \rightarrow [Cu(NH_3)_4]^{+2}$

(2) $Cu^{+2} + 4CN^{-} \rightarrow [Cu(CN)_4]^{-2}$

(3) $Cu^{+2} + 2en \rightarrow [Cu(en)_2]^{+2}$

(4) $Cu^{+2} + 4H_2O \rightarrow [Cu(H_2O)_4]^{+2}$

38) The compounds $[Co(SO_4)(NH_3)_5]Br$ and $[Co(SO_4)(NH_3)_5]Cl$ represent :-

(1) Linkage isomerism

(2) Ionisation isomerism

(3) Coordination isomerism

(4) no isomerism

39) Among the following transition metal ions, the one where all metal ions have 4 unpaired electrons:

(1) Cr^{+2}, Mn^{+2}, Fe^{+2}

(2) Cr^{+2}, Mn^{+3}, Fe^{+2}

(3) Mn^{+3}, Fe^{+3}, Co^{+3}

(4) Fe^{+2}, Co^{+3}, Ni^{+2}

40) Given below are two statements :
Statement-I : Aqueous solution of $K_2Cr_2O_7$ is preferred as a primary standard in volumetric analysis over $Na_2Cr_2O_7$ aqueous solution.
Statement-II : $K_2Cr_2O_7$ has a higher solubility in water than $Na_2Cr_2O_7$.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Both statement I and statement II are false

(4) Statement I is true but statement II is false

41) When excess of KCN is added into $CuSO_4$ solution then product obtained is

(1) $Cu(CN)_2$

(2) CuCN

(3) $K_2[Cu(CN)_2]$

(4) $K_3[Cu(CN)_4]$

42) Which of the following pairs of isomers and types of isomerism are correctly matched?
(I) $[Co(NH_3)_5(NO_2)]Cl_2$ and $[Co(NH_3)_5(ONO)]Cl_2$ ..... Linkage
(II) $[Cu(NH_3)_4][PtCl_4]$ & $[Pt(NH_3)_4][CuCl_4]$ ..... Coordinations
(III) $[Pt(NH_3)_4Cl_2]Br_2$ & $[Pt(NH_3)_4Br_2]Cl_2$ ..... Ionization
Select the correct answer using the codes given below :

(1) 1 and 2

(2) 2 and 3

(3) 1 and 3

(4) 1, 2 and 3

43) The molecular shapes of diborane is shown : Consider the following statements for diborane :
1. Boron is approximately sp^3 hybridized
2. B–H–B angle is 180°
3. There are two terminal B–H bonds for each boron atom
4. There are only 12 bonding electrons
Available of these statements :

(1) 1, 3 and 4 are correct

(2) 1, 2 and 3 are correct

(3) 2, 3 and 4 are correct

(4) 1, 2 and 4 are correct

44) Which one of the following complexes has the largest C. F. S. E. (crystal field stabilization energy)?

(1) $[Mn(H_2O)_6]^{2+}$

(2) $[Cr(H_2O)_6]^{2+}$

(3) $[Mn(H_2O)_6]^{3+}$

(4) $[Cr(H_2O)_6]^{3+}$

45) Which of the following is not a bidentate ligand?

(1) Acetyl acetonate (acac)

(2) NH_2–NH_2

(3) oxalate (ox)

(4) dimethyl glyoximate

BIOLOGY

1) What is true for Bt-toxin:-

(1) Bt-toxin protein exists as active toxin in bacterium

(2) Bt-toxin is converted into an active toxin due to acidic pH in gut of insect

(3) Bt-toxin is converted into an active toxin due to alkaline pH in gut of insects

(4) Bt-toxin is formed by *Trichoderma*

2) Fill up the blanks.
I. Using *Agrobacterium* vectors, ...A.... specific genes are introduced into the host plant.
II. RNAi takes place in all eukaryotic organisms as a method of .....B.....
III. The genes .....C.... and .....D.... control cotton bollworms.
IV. ....E... has been introduced in Bt corn to protect the same from corn borer.
A to E in above statements can be:-

(1) A-fungal, B-pest resistance,
C-*cryIIAb*, D-*cryIAb*, E-*cryIAc*

(2) A-nematode, B- Cellular defence
C-*cryIAc*, D-*cryIIAb*, E-*cryIAb*

(3) A-insects, B-cellular defense,
C-*cryIIAb*, D-*cryIAc*, E-*cryIAb*

(4) A-virus, B-protection against viral infection,
C-*cryIAc*, D-*cryIIAb*, E-*cryIIAc*

3) Read the following statement and choose correct option :-
I. Recombinant DNA technology is used to improve crop plants by increasing their productivity, by making them more nutritious and by developing disease resistance
II. Bt cotton is resistant to bollworm infestation
III. *Bacillus thuringiensis* is not harmed by self Cry protein because of its occurrence as protoxin (inactive)
IV. Protoxin Cry protein is changed into active Cry protein in the stomach of insects due to acidic pH in stomach

(1) All are correct

(2) I and IV are correct

(3) Only IV is false

(4) All are false

4)

Read the following statements :-
(a) Nematode resistant plant is developed by novel strategy based on process of RNA interference (RNAi).
(b) RNAi takes place in all eukaryotic organisms as method of cellular defense.
(c) RNAi involves silencing of specific mRNA due to complementary dsRNA.

(d) Transgenic host expressing two non-complementary RNAs that initiate RNAi.
Correct statements are :

(1) a, b

(2) a, b, d

(3) a, b, c

(4) a, b, c, d

5)

Which of the following correctly defines a transgenic animal:-

(1) An animal which has foreign DNA and RNA in some of its cells because of an injection of DNA and RNA into the nucleus of the zygote from which it is developed

(2) An animal which has foreign DNA in all its cells because of an injection of DNA into the nucleus of the zygote from which it is developed

(3) An animal which has foreign DNA in some of its cells because of an injection of DNA into the nuclei of some of the cells of the blastocyst

(4) An animal which has foreign DNA in all its cells because of an injection of DNA into the nuclei of some of the cells in adulthood

6) The illegal and unlawful use of biomaterials without payment to the inhabitants of their origin is called :

(1) Biopatent

(2) Biotechnology

(3) Bio-war

(4) Biopiracy

7)

Which of the following is wrongly matched in the given table ?

(i)	Golden rice	Transgenic rice	Vitamin A enriched
(ii)	*Flavr savr*	Transgenic tobacco	Delayed ripening
(iii)	Bt-cotton	Transgenic cotton	Bacterial and viral resistance
(iv)	Rosie	Transgenic cow	Milk contained human alpha lactoglobulin

(1) i and ii

(2) ii, iii and iv

(3) Only ii and iii

(4) Only iii

8) Indian government organisation which make decision regarding validity of GM research and safety of introducing GMO for public services is:-

(1) IARI

(2) GEAC

(3) CDFD

(4) Eli Lilly

9) First transgenic cow *Rosie* milk is rich in which protein:

(1) Human alpha globin

(2) Human alpha lactalbumin

(3) Human beta lactalbumin

(4) Human beta globin chain

10) Which of the following statement is true regarding RNA interference?

(1) RNAi takes place in all living organisms as a method of cellular defense.

(2) RNAi involves silencing of a specific DNA due to a single stranded RNA

(3) Source of complementary RNA could be from an infection by RNA viruses or transposons that replicate via RNA intermediate.

(4) Using pBR322 vector, nematode specific genes were introduced into the host plant.

11) Human insulin as a prohormone contains-

(1) A, B & C Peptide

(2) C Peptide only

(3) A & B peptide only

(4) B & C Peptide only

12) Which of the following is not the use of transgenic animal ?

(1) Study of normal physiology and development

(2) Testing of vaccine safety

(3) Production of biological products

(4) Molecular diagnosis

13) ELISA is :-

(1) Technique of disease diagnosis based on PCR

(2) Technique of gene therapy based on antigen-antibody interaction.

(3) Technique of molecular diagnosis based antigen-antibody interaction

(4) Technique of conventional diagnosis

14) The technique which is now routinely used to detect HIV in suspected AIDS patient is :-

(1) Serum analysis

(2) Urine analysis

(3) PCR

(4) Widal test

15) Which of the following is not an incorrect statement ?

(1) In India GEAC will make decisions regarding the validity of GM research and safety of introducing GM-organism for public services.

(2) Golden rice is distinct for its unique aroma and flavour and 27 varieties of golden rice are grown in India.

(3) The Indian Parliament did not do any amendment in the Indian Patent Bill that explain biotechnological issues.

(4) In 1897 the first transgenic cow, Rosie was developed which produced milk rich in vit B_{12}.

16) **Statement-A :** At present about 30 recombinant therapeutics have been approved for human use the world over.
Statement-B : 27 documented varieties of Basmati rice are grown in India.

(1) Both A and B are true

(2) Only A is true

(3) Only B is true

(4) Both A and B are false

17) **Statement I :-** ADA enzyme is transferred to patient for permanent cure.
Statement II :- ADA deficiency can be permanently cured by ADA enzyme replacement therapy and bone marrow transplantation.

(1) Only Statement I is correct

(2) Only Statement II is correct

(3) Both Statements are correct

(4) Both Statements are incorrect

18) **Assertion :** Biofortification is used to increase nutrient value of crops.
Reason : Meristem culture is used to obtain virus free plants.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

19) Match the column-I with column-II and select the correct options :–

Column-I		**Column-II**	
(i)	First clinical gene therapy	(a)	1990
(ii)	First made recombinant DNA	(b)	1972
(iii)	Eli-Lilly formed Human insulin	(c)	1983
(iv)	American company got patent rights on Basmati rice	(d)	1997

(1) i – a, ii – c, iii – b, iv – d

(2) i – a, ii – b, iii – c, iv – d

(3) i – c, ii – b, iii – d, iv – a

(4) i – c, ii – b, iii – a, iv – d

20) *Agrobacterium tumefaciens* is known as :-

(1) Natural genetic engineer of plant

(2) Natural genetic engineer of animal

(3) Symbiotic bacteria

(4) Symbiotic plant

21) Which of the following enzymes has been incorrectly matched with their function?

(1) Ligase-Molecular glue

(2) Endonuclease-Molecular scissor

(3) DNA polymerase-Joins nucleosides

(4) RNA polymerase- Joins nucleotides

22) Which of the following pair is incorrect ?

(1) Microinjection — animal cell

(2) Biolistic method — plant cell

(3) Plasmid — Virus

(4) EcoR I — restriction endonuclease enzyme

23) Choose the correct matches :-

(a)	First Restriction Endonuclease	Hind II
(b)	EcoR I and EcoR V	Sticky ends producing endonuclease
(c)	cDNA	Shorter than original gene
(d)	Direct method of gene transfer	*Agrobacterium* mediated.
(e)	Conventional methods of diagnosis	Serum and urine analysis
(f)	PCR	RNA primers

(1) a, c, f

(2) b, d, f

(3) a, c, e

(4) b, c, e

24) Fill up the blanks. (A) When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of _____ and these can be joined together for end-to-end using _____. (B) Disruption of the cell wall can be achieved by treating the bacterial cells, plant or animal tissue with enzymes such as _____ (bacteria), _____ (plant cells) and _____ (fungus).
(C) If any protein encoding gene is expressed in a heterologous host, it is called as _____. (D) In gel electrophoresis, the separated DNA fragments can be seen only after staining the DNA with a compound known as _____ followed by exposure to UV radiation.

(1) A = sticky ends, recombinase; B = lysozyme, cellulase, chitinase;
C = recombinant protein; D = ethidium bromide

(2) A = sticky ends, DNA ligases; B = exonuclease, cellulase, chitinase;
C = recombinant protein; D = ethidium bromide

(3) A = cut ends, DNA topoisomerase; B = hemicellulose, cellulase, chitinase; C = native glycoprotein; D = CsCI

(4) A = sticky ends, DNA ligases; B = lysozyme, cellulase, chitinase; C = recombinant protein; D = ethidium bromide

25) What happens when DNA fragment is inserted in to plasmid pBR322 using enzyme Pvu I or Pst I ?
(a) Inactivation of amp^R gene
(b) Inactivation of tet^R gene
(c) Unable to grow in presence of ampicillin
(d) Unable to grow in presence of tetracycline
(e) Unable to replicate

(1) b & d are correct

(2) a and c are correct

(3) a, c & e are correct

(4) b, d & e are correct

26) Read the following four statements (A-D) :-
(A) The first transgenic buffalo, Rosie produced milk which was human alpha-lactalbumin enriched.
(B) Restriction enzymes are used in isolation of DNA from other macro molecules.
(C) Downstream processing is one of the steps of r-DNA technology.
(D) Disarmed pathogen vectors are also used in transfer of r-DNA into the host.
Which two statement are wrong ?

(1) Statements (A) and (B)

(2) Statements (B) and (C)

(3) Statements (C) and (D)

(4) Statements (A) and (C)

27) If the plasmid in the bacteria does not have any insert then the colonies produce :-

(1) Blue colour in the presence of X-gal

(2) No colour in the presence of X-gal

(3) Blue colour in the absence of X-gal

(4) Blue colour in the presence of maltose

28) β-galactosidase, Lysozyme, Amylase, Maltase, Chitinase, Cellulase, Transacetylase.
How many of the above enzymes are used during isolation of DNA for genetic engineering in various organisms?

(1) One

(2) Two

(3) Three

(4) Four

29) The method in which alien DNA is directly injected into the nucleus of an animal cell is known as :-

(1) Biolistic

(2) Micro-injection

(3) Electrophoresis

(4) Heat-shock treatment

30) Select the correct match :

(1)	Insertional inactivation	–	Differentiate recombinants from non recombinants
(2)	Recombinant protein	–	Any protein encoding gene is expressed in a homologous host.
(3)	Downstream processing	–	A series of process which start from raw material and to completion of biosynthetic product
(4)	Cloning	–	Introduction of the identified DNA into the host.

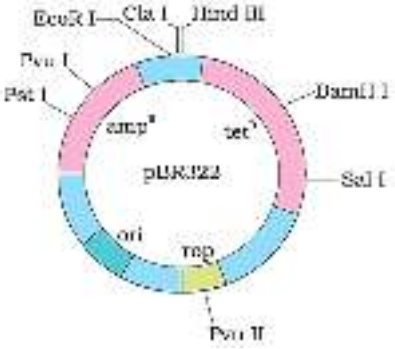
(1) 1

(2) 2

(3) 3

(4) 4

31) The figure below is the diagrammatic representation of the *E.coli* vector pBR322. Which one of



the given options correctly identifies its certain component(s) ?

(1) BamH I, EcoRI - Selectable marker

(2) Ori - Original restriction enzyme

(3) rop - Codes for the protein involved in the replication of plasmid

(4) amp^R, tet^R - site from where replication starts

32) Select the incorrect statement for continuous culture system :-

(1) In this used medium is drained out from one side while fresh medium is added from other side.

(2) In this cells are maintained in their physiologically most active lag phase of growth.

(3) It produces larger biomass

(4) It shows higher yields of desired product.

33)

Arrange the steps of rDNA technology in correct order:-
I. Extraction of the desired gene product.
II. Amplification of gene of interest.
III. Isolation of desired DNA fragment.

IV. Ligation of DNA fragment into vector.

V. Insertion of rDNA into host.

(1) I, II, III, IV, V

(2) V, IV, III, II, I

(3) III, II, IV, V, I

(4) III, IV, II, I, V

34) A typical bioreactor has :
(a) An agitator system
(b) An oxygen delivery system
(c) A foam control system
(d) A temperature control system
(e) A pH control system
(f) Sampling ports

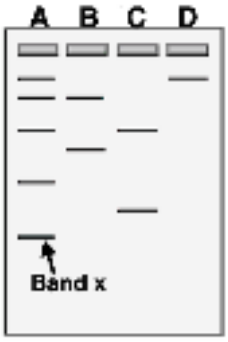
(1) a, b and c

(2) a, b, c and d

(3) a, b, c, d and e

(4) a, b, c, d, e and f

35) DNA samples were collected from four organisms A, B, C and D. The diagram below represents



the results of a procedure that separated the DNA in each sample What does band 'X' represent ?

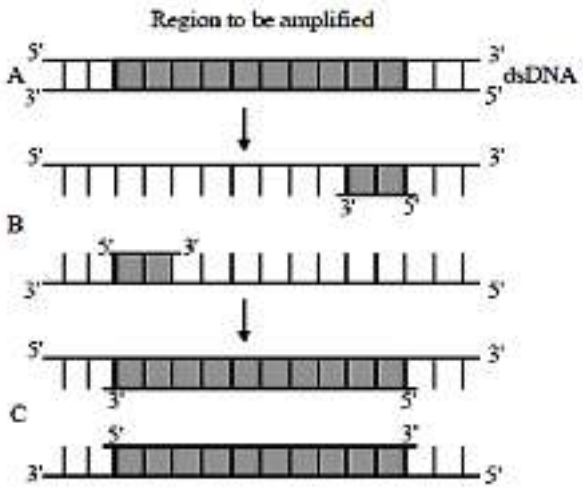
(1) Largest fragment of DNA that travelled the fastest

(2) Largest fragment of DNA that travelled the slowest

(3) Smallest fragment of DNA that travelled the slowest

(4) Smallest fragment of DNA that travelled the fastest

36) The given below figure shows three steps (A,B,C) in Polymerase Chain Reaction (PCR). Select the option having correct identification with any one representation.



(1) C-Extension in the presence of heat stable DNA polymerase

(2) A-Annealing with two sets of primers

(3) B-Denaturation at a temperature of about 98°C separating the two DNA strands

(4) A-Denaturation at a temperature of about 50°C

37) **Statement I :-** DNA is a negatively charged molecule, hence it moves towards the positive electrode (anode).
Statement II:- In a continuous culture system, the cells are maintained in their physiologically most active lag phase.

(1) Statement I and II both are correct.

(2) Statement I and II both are incorrect.

(3) Only Statement I is correct.

(4) Only Statement II is correct.

38) **Statement I :-** In gel electrophoresis, the separated DNA fragments can be visualised only after staining the DNA with ethidium bromide followed by exposure to visible light.
Statement II :- The most commonly used matrix in gel electrophoresis is agarose.

(1) Statement I and II both are correct.

(2) Statement I and II both are incorrect.

(3) Only Statement I is correct.

(4) Only Statement II is correct.

39) **Assertion :** Agarose gel electrophoresis is employed to check the progression of restriction enzyme digestion.
Reason : Both the passenger and vehicle DNAs are cleaved by using same restriction endonuclease enzyme.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

40) **Assertion** :- The activated toxin binds to the surface of midgut epithelial cells & create pores that cause cell swelling, lysis & finally death of insect. **Reason** :- Most of Bt toxins are insect group specific.

(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

41) **Statement I :** An individual may have births and deaths, but a population has birth rates and death rates.
Statement II : In a population birth rates and death rates refer to per capita births and deaths.

(1) Statement I is correct and statement II is incorrect.

(2) Statement I is incorrect and statement II is correct.

(3) Both statement I and statement II are correct.

(4) Both statement I and statement II are incorrect.

42) **Statement I :** The size of a population for any species is a static parameter.
Statement II : The Size of a population keeps changing with time, depending on various factors including food availability, predation pressure, and adverse weather.

(1) Statement I is correct, and Statement II is incorrect.

(2) Statement I is incorrect, and Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both statement I and statement II are incorrect.

43) **Statement I :** Natality refers to the number of births during a given period in the population that are added to the initial population density.
Statement II: Mortality is the number of deaths in the population during a given period.

(1) Statement I is correct, and Statement II is incorrect.

(2) Statement I is incorrect, and Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both statement I and statement II are incorrect.

44) **Statement I :** Changes in population density give us some ideas of what is happening to the population, whether it is flourishing or declining.
Statement II : The density of a population in a given habitat during a given period fluctuates due to changes in three basic processes, i.e, natality, mortality, and immigration only.

(1) Statement I is correct and statement II is incorrect.

(2) Statement I is incorrect and statement II is correct.

(3) Both statement I and statement II are correct.

(4) Both statement I and statement II are incorrect.

45) **Statement I :** If N is the population density at time t, then its density at time t +1 is $N_{t+1} = N_t +$

[(B + I) + (D + E)].
Statement II : Population density will increase if the number of births plus the number of immigrants (B + I) is more than the number of deaths plus the number of emigrants (D + E).

(1) Statement I is correct and statement II is incorrect.
(2) Statement I is incorrect and statement II is correct.
(3) Both statement I and statement II are correct.
(4) Both statement I and statement II are incorrect.

46) The ecological principle behind biological control of pest insects is :

(1) Mutualism
(2) Predation
(3) Proto-cooperation
(4) Competition

47) Which of the following statements correctly differentiates population from community?

(1) Population is a group of similar individuals, community is a group of populations of different species.
(2) Population is all organisms in a habitat, community is only plants.
(3) Population is only producers, community is only consumers.
(4) Population and community are the same.

48) Which of the following pairs is incorrectly matched?

(1) Commensalism – Orchid on mango tree
(2) Parasitism – Cuscuta on hedge plants
(3) Camouflage – Frog blending with green leaves
(4) Mutualism – Tiger and deer

49) Which of the following is not a characteristic of a population?

(1) Natality
(2) Mortality
(3) Age distribution
(4) Hormonal secretion

50) **Assertion (A):** Plants produce alkaloids, latex, and thorns to prevent herbivory.
Reason (R): These adaptations make plants more nutritious for herbivores.

(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

51) In logistic growth, population size becomes stable at:

(1) Doubling time

(2) Carrying capacity

(3) Intrinsic rate of increase

(4) Lag phase

52) **Assertion (A):** In Connell's field experiment, *Chathamalus* survived better in the absence of *Balanus*.
Reason (R): *Balanus* excludes *Chathamalus* from the lower intertidal zone by competition.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

53) Which of the following interactions is most likely to promote coexistence of competing species?

(1) Competitive exclusion

(2) Resource partitioning

(3) Predation

(4) Amensalism

54) **Assertion (A):** Predator–prey relationship contributes to maintaining species diversity in a community.
Reason (R): Predators keep prey populations under control, preventing competitive exclusion.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

55) Which of the following is not an adaptation of desert plants?

(1) Thick cuticle

(2) Sunken stomata

(3) Large, thin leaves

(4) Photosynthetic stems

56) **Assertion (A):** Primary productivity depends on the plant species inhabiting a particular area.
Reason (R): It is measured in terms of kcal m^{-2} yr^{-1}.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

57) Identify the correct statements:

a. In a stable ecosystem, nutrient cycling and energy flow remain balanced.
b. A forest ecosystem shows greater resistance to environmental disturbances than a crop field.
c. Resistance and resilience are features of population growth, not of ecosystems.
d. Ecosystem stability is related to species diversity.

(1) a, b, and d only

(2) a and c only

(3) b and c only

(4) a and d only

58)

Match the Following

A.	Detritivores	i.	Break down detritus into smaller particles
B.	Decomposers	ii.	Convert inorganic nutrients into usable form
C.	Humification	iii.	Formation of a dark-coloured layer of partially decomposed organic matter
D.	Mineralisation	iv.	Release of inorganic nutrients

(1) A–i, B–ii, C–iii, D–iv

(2) A–ii, B–i, C–iv, D–iii

(3) A–iii, B–ii, C–i, D–iv

(4) A–iv, B–iii, C–ii, D–i

59) Match the Following -

A.	Commensalism	i.	Sea anemone and clownfish
B.	Parasitism	ii.	*Cuscuta* on *hibiscus*
C.	Mutualism	iii.	Mycorrhiza
D.	Competition	iv.	Flamingos and fishes in the same lake

(1) A–i, B–ii, C–iii, D–iv

(2) A–ii, B–i, C–iii, D–iv

(3) A–iii, B–ii, C–iv, D–i

(4) A–iv, B–iii, C–ii, D–i

60)

Select the incorrect statements:
a. Gross primary productivity is the rate of formation of organic matter after respiration losses.
b. Secondary productivity refers to biomass produced by consumers.
c. Net primary productivity is available to herbivores.
d. Decomposers help in mineral cycling.

(1) a only

(2) b and c

(3) a and b

(4) d only

61) Correct statements are standing crop and standing state are—
a. Standing crop is the mass of living material at a given time.
b. Standing state is the quantity of nutrients present in the soil or water.
c. Both are part of an ecosystem's structure.
d. Standing crop is expressed only in energy units.

(1) a, b, c only

(2) b and d only

(3) a and c only

(4) a, b, c, and d

62) A hypothetical population of birds shows very high natality, very low mortality, and abundant food. The growth curve will be:

(1) Sigmoid (logistic)

(2) Exponential (J-shaped)

(3) Constant

(4) Declining

63) Which of the following pairs correctly represents a keystone species and its role in the ecosystem?

(1) Tiger – maintains herbivore population in forest

(2) Bacteria – Cheif producers in soil

(3) Grass – primary producer

(4) Goat – herbivore maintaining vegetation

64) Researchers compared two grassland plots:
• Plot A had 15 species of grasses.
• Plot B had 5 species of grasses.
After a drought, Plot A recovered quickly, but Plot B did not.
Which conclusion aligns with NCERT's principle?

(1) Lesser diversity ensures higher productivity.

(2) Diversity ensures ecosystem stability.

(3) High diversity reduces stability.

(4) Stability and diversity are unrelated.

65) In a controlled experiment, plants were grown in two tanks:
• Tank X: decomposers present
• Tank Y: decomposers removed
After several weeks, nutrient levels in Tank Y dropped drastically.
What can be concluded?

(1) Producers can recycle nutrients without decomposers.

(2) Decomposers are essential for nutrient cycling.

(3) Nutrient cycling depends only on sunlight.

(4) Decomposers reduce soil fertility.

66) **Statement 1:** Keystone species have a low functional impact despite their large population.
Statement 2: Removal of a keystone species can destabilize the entire ecosystem.

(1) Both statements are true, and Statement 2 correctly explains Statement 1.

(2) Both statements are true, but Statement 2 does not explain Statement 1.

(3) Statement 1 is false, Statement 2 is true.

(4) Both statements are false.

67) Two leaf litters, pine and banyan, were compared. After one month, banyan litter decomposed faster.
Which factor most likely explains this difference?

(1) Pine litter contains less cellulose and more nitrogen.

(2) Banyan litter is richer in lignin and decomposes slowly.

(3) Banyan litter contains more nitrogen and less lignin.

(4) Pine litter is easily decomposed by microbes.

68) In a pond ecosystem, a increase in detritus-feeding bacterial population occurred after heavy leaf fall. This population rise represents:

(1) Logistic growth

(2) Decline phase

(3) Zero population growth

(4) Seasonal migration

69)

Match the Following

A.	Fragmentation	i.	Breakdown of detritus into small particles by detritivores
B.	Leaching	ii.	Conversion of complex organic matter into inorganic forms
C.	Mineralisation	iii.	Water-soluble nutrients lost from detritus
D.	Humification	iv.	Formation of dark, amorphous, stable organic matter

(1) A–i, B–iii, C–ii, D–iv

(2) A–iii, B–iv, C–ii, D–i

(3) A–ii, B–i, C–iii, D–iv

(4) A–i, B–iv, C–iii, D–ii

70) High plant diversity in tropical forests enhances decomposition because:

(1) It increases detritus quantity and microbial diversity.

(2) It reduces nutrient recycling efficiency.

(3) It limits herbivore population.

(4) It lowers soil respiration.

71) Match the following and select the correct option:

	Column I		Column II
A.	Produce a large number of small-sized offspring	I.	Birds
		II.	Pelagic fishes
B.	Produce a small number of large-sized offspring	III.	Oyster
		IV.	Mammals

(1) A–I, II ; B–III, IV

(2) A–I, III ; B–II, IV

(3) A–II, III ; B–I, IV

(4) A–I, IV ; B–II, III

72)

Read the following statements regarding population growth and select the correct one:

(1) Exponential growth continues under limited resources.

(2) Sigmoid curve lacks deceleration phase.

(3) Logistic growth shows lag, exponential, deceleration and asymptote.

(4) Carrying capacity always increases.

73) **Assertion (A):** Conservation of tropical rainforests is crucial for maintaining global biodiversity.
Reason (R): Tropical rainforests receive high annual rainfall.

(1) Both A and R are true and R explains A

(2) Both A and R are true but R does not explain A

(3) A true, R false

(4) A false, R true

74) **Assertion (A):** Predators help maintain species diversity.
Reason (R): Invasive weeds cause environmental damage.

(1) Both true, R explains A

(2) Both true, R does not explain A

(3) A true, R false

(4) A false, R true

75) Select correct statements:
A. In current mass extinction, Extinction rates are 100–1000 times lower.
B.In last 500 years, ~700 species extinct, >15,500 threatened.
C. Growth limited by carrying capacity.

D. Biodiversity loss reduces growth rate.
E. Biodiversity loss affects only function.

(1) A, B, C
(2) B, C, D
(3) C, D, E
(4) A, D, E

76) Which statements are correct?
A. Amazon has highest biodiversity
B. Decomposition faster in tropics
C. Biodiversity affects productivity

(1) A and B
(2) B and C
(3) A and C
(4) All of these

77) **Assertion (A):** Biodiversity is result of evolution.
Reason (R): Mutation and recombination create variation.

(1) Both true, R explains A
(2) Both true, R does not explain A
(3) A true, R false
(4) A false, R true

78) Identify the odd one with respect to biodiversity conservation.

(1) Cryopreservation
(2) Gene banks
(3) Cross breeding
(4) Seed banks

79) Match biodiversity level:
A. Genetic — 1. Rice varieties
B. Species — 2. Mammals
C. Ecosystem — 3. Forest/desert

(1) A–1, B–2, C–3
(2) A–2, B–3, C–1
(3) A–3, B–1, C–2
(4) A–1, B–3, C–2

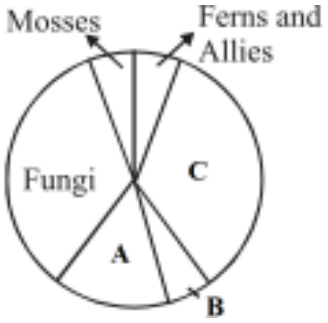
80) Biotechnology supports biodiversity by:

(1) Cloning invasive species
(2) Increasing uniformity

(3) Cryopreservation

(4) GM predators

81)



Above diagram shows proportionate number of species of major taxa of plants, identify the A, B, C plant group :-

	A	B	C
(1)	Lichens	Angiosperms	Algae
(2)	Angiosperms	Algae	Lichens
(3)	Lichens	Algae	Angiosperms
(4)	Algae	Lichens	Angiosperms

(1) 1

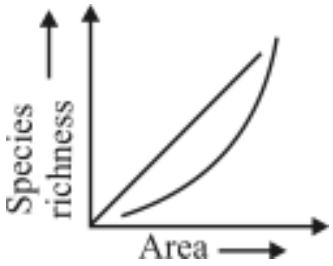
(2) 2

(3) 3

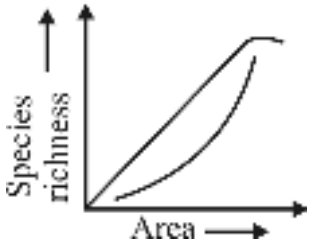
(4) 4

82) Identify species area relationship graph :

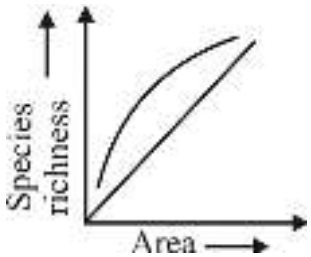
(1)



(2)



(3)



(4) 

83) In the given diagram, which of the following option correctly represent A, B, C and D regarding biodiversity of invertebrates ?

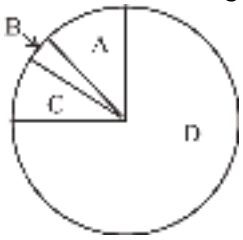


Figure :- Representing global biodiversity, Proportionate number of species of major taxa of Invertebrates.

	A	B	C	D
(1)	Other animal groups	Molluscs	Crustaceans	Insects
(2)	Other animal groups	Crustaceans	Molluscs	Insects
(3)	Insects	Crustaceans	Molluscs	Other animal groups
(4)	Insects	Molluscs	Crustaceans	Other animal groups

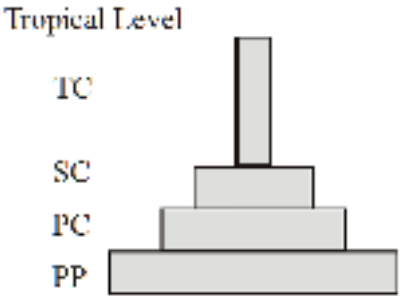
(1) 1

(2) 2

(3) 3

(4) 4

84) Given pyramid is observed in :-



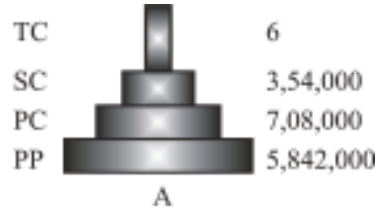
(1) Pyramid of number without any exception

(2) Pyramid of energy without any exception

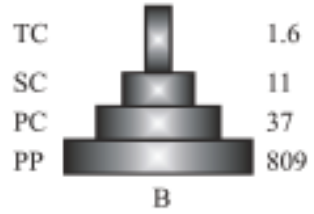
(3) Pyramid of biomass without any exception

(4) Both (1) and (3)

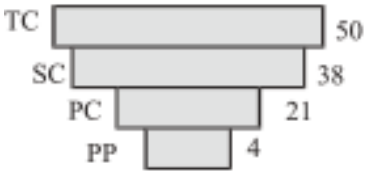
85) Study the following ecological pyramids carefully :-



A



B



C

Match the following statements (i), (ii) and (iii) with given pyramids and select the correct answer.

(i) Inverted pyramid of biomass depicting small standing crop of phytoplanktons supporting a large standing crop of zooplanktons.

(ii) Pyramid of numbers in a grassland ecosystem showing about 6 million producers.

(iii) Upright pyramid of biomass.

(1) A-(ii), B-(iii), C-(i)

(2) A-(ii), B-(i), C-(iii)

(3) A-(i), B-(iii), C-(ii)

(4) A-(i), B-(ii), C-(iii)

86) The given diagram is related to which of the species association?



(1) Mutualism

(2) Commensalism

(3) Proto co-operation

(4) Ammensalism

87) Which plant is pollinated by insect in figure given below ? 

(1) Orchid flower

(2) Fig flower

(3) Sunflower

(4) Lotus flower

88) With respect to Tilman's outdoor plots experiment, identify the correct statement from the below statements.

(1) Plots with more species show less year to year variation in total biomass.

(2) Decreased diversity contributed to higher productivity.

(3) Increased diversity contributed to lower productivity.

(4) Plots with less species show less year to year variation in total biomass.

89) Read the given statements and select the correct option regarding this.
(i) Ecosystem services are provided by nature to human beings such as oxygen for respiration, aesthetic value, etc.
(ii) Direct economic benefits derived from nature by human beings such as food, medicines, etc.
(iii) Every species has an intrinsic value, even if it is not of any economic use to us.
With respect to above given codes (i), (ii) and (iii) choose the correct option.

	Narrowly utilitarian	**Broadly utilitarian**	**Ethical**
(1)	(iii)	(ii)	(i)
(2)	(ii)	(i)	(iii)
(3)	(i)	(ii)	(iii)
(4)	(ii)	(iii)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

90) **Assertion:** Amazon rain forest are called lungs of the plant.
Reason: The 'Evil Quartet' is a sobriquet related to the biodiversity loss.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	1	2	4	3	3	3	3	4	3	1	2	3	4	3	4	2	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	1	1	1	3	2	1	4	3	2	1	3	4	1	3	2	4	1	1	2
Q.	41	42	43	44	45															
A.	1	4	1	1	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	1	2	4	4	4	2	2	4	3	4	3	1	2	3	4	1	4	1	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	1	1	1	1	1	4	2	4	1	2	2	2	4	3	2	2	4	2	4
Q.	86	87	88	89	90															
A.	4	4	1	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	3	3	2	4	2	2	2	3	1	4	3	3	1	1	4	2	2	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	3	4	2	1	1	3	2	1	3	2	3	4	4	1	3	4	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	3	1	2	2	1	4	4	3	2	1	2	1	3	2	1	1	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	2	1	2	2	3	3	1	1	1	3	3	2	2	2	4	1	3	1	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	3	2	2	1	1	1	1	2	2										

SOLUTIONS

PHYSICS

1) **Explaining :** To comment on the brightness of bulbs.
Concept : This question is based on power
Solution :
For B_1 branch

$$i_1 = \frac{V}{\sqrt{R^2 + \left(\frac{1}{2\pi fc}\right)^2}}$$

If f↑ ⇒ i_1↑ ⇒ Brightness↑
for B_2 branch

$$i_2 = \frac{V}{\sqrt{R^2 + (2\pi fL)^2}}$$

If f↑ ⇒ i_2↓ ⇒ Brightness↓
Final Answer : (3)

2) **Explaining :** In given circuit applied voltage V = $50\sqrt{2}$ sin (100πt) volt and ammeter reading is 2A then calculate value of L?
Concept : This question is based on $X_L = L\omega$.
Solution : $V_{rms} = I_{rms} X_L$ ∵ Reading of ammeter = I_{rms}

$$X_L = \frac{V_{rms}}{I_{rms}} = \frac{V_0}{\sqrt{2}\, I_{rms}} = \frac{50\sqrt{2}}{\sqrt{2} \times 2} = 25\ \Omega$$

$$L = \frac{X_L}{\omega} = \frac{25}{100\pi} = \frac{1}{4\pi}\text{H}$$

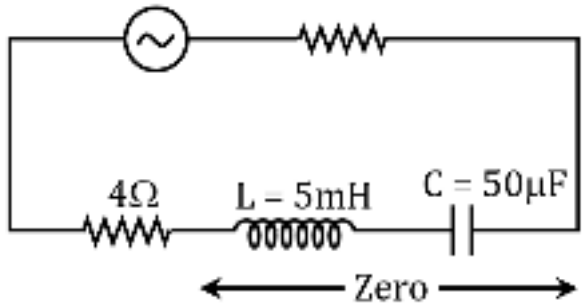
Final Answer : (1)

3) **Explaining :** In the circuit shown below, the ac source has voltage V = 20 cos(ωt) volts with ω = 2000 rad/sec. reading of AC voltmeter b
Concept : This question is based on $Z = \sqrt{R^2 + (X_L - X_C)^2}$.
Solution : R = 6 + 4 = 10Ω
$X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10\Omega$

$$X_C = \frac{1}{\omega C} = \frac{1}{2000 \times 50 \times 10^{-6}} = 10\Omega$$

$$\therefore\ Z = \sqrt{R^2 + (X_L - X_C)^2} = 10\Omega$$

Amplitude of current = $i_0 = \frac{V_0}{Z} = \frac{20}{10} = 2\text{A}$
Since the circuit is under resonance, voltage across L–C combination is zero.



So, Voltmeter reading = Voltage across 4Ω

$$= \left(\frac{2}{\sqrt{2}}\right)(4) = 5.6V$$

Final Answer : (1)

4) **Explaining :** V = 200sin $\left[\omega t + \frac{\pi}{6}\right]$ and $I = 10 \sin\left[\omega t + \frac{\pi}{3}\right]$ calculate power

Concept : This question is based on P = V_{rms} I_{rms} cosϕ.

Solution : $\phi = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6} = 30^\circ$

p = V_{rms} I_{rms} cos30°

$= \frac{200}{\sqrt{2}} \times \frac{10}{\sqrt{2}} \cos 30^\circ$

$= 500\sqrt{3}W$

Final Answer : (2)

5) **Explanation:** To match different AC circuits with their characteristic phase relationships.

Concept Involved: The phase difference between voltage and current is determined by the components in the circuit. In an RL circuit, voltage leads current. In an RC circuit, current leads voltage. In a purely inductive circuit, voltage leads current by 90°. In a purely resistive circuit, voltage and current are in phase (0°).

Solution:

(A) RL circuit: It is an inductive circuit, so voltage leads the current. Matches with (2).

(B) RC circuit: It is a capacitive circuit, so current leads the voltage. Matches with (1).

(C) Pure Inductive circuit: The phase difference between voltage and current is exactly 90°, with voltage leading current. Matches with (4).

(D) Pure Resistive circuit: The phase difference between voltage and current is 0°, as they are in phase. Matches with (3). The correct match is (A)→(2), (B)→(1), (C)→(4), (D)→(3).

Final Answer: (4) (A)→(2); (B)→(1); (C)→(4) ; (D)→(3)

6) **Explaining :** To find rms value of current.

Concept : This question is based on RMS value ; $I_{rms}^2 = \langle i^2 \rangle$

Solution : $i_{rms} = \sqrt{\frac{1}{T}\int_0^T i^2 dt} = \frac{T^2}{\sqrt{5}}$

Final Answer : (3)

7) **Explaining :** To find capacitance of circuit

Concept : This question is based on power factor

Solution :

p.f. = $\cos\phi = \frac{1}{\sqrt{2}} = \frac{R}{Z}$

$\Rightarrow Z = \sqrt{2}R \Rightarrow Z = 10\sqrt{2}\Omega$

Here $Z = \sqrt{R^2 + (X_C - X_L)^2}$

where X_L = = ωL = (100)(0.1) = 10 Ω

$200 = 100 + (X_C - 10)^2$

$\Rightarrow X_C = 20\Omega$

$\Rightarrow \frac{1}{100C} = 20$

$\Rightarrow$ C = 500 µF

Final Answer : (3)

8) **Question Explanation:**

Complete electrical analysis of a series LC circuit to verify current and individual component voltages.

Concept:

Series LC Circuit Parameters.

In a series circuit where Resistance (R) is negligible:

Net Reactance (X): $X = |X_C - X_L|$

RMS Current (I): $I = \frac{V_{rms}}{X}$

Voltage across Inductor (V_L): $V_L = I \cdot X_L$

Voltage across Capacitor (V_C): $V_C = I \cdot X_C$

Solution:

Calculate Total Reactance (X):

Given $X_L = 25\Omega$ and $X_C = 75\Omega$.

$X = |75\Omega - 25\Omega| = 50\Omega$

Calculate Current (I):

The source voltage is $250V$.

$I = \frac{250V}{50\Omega} = 5A$

Verifying the Options based on $I = 5A$:

Voltage across Inductor (V_L): $5A \times 25\Omega = 125V$

Voltage across Capacitor (V_C): $5A \times 75\Omega = 375V$

Final Answer: Option (3)

9) **Explaining :** To find the frequency of AC source

Concept : This question is based on choke coil

Solution :

With dc : $P = \frac{V^2}{R} \Rightarrow R = \frac{(10)^2}{20} = 5\Omega;$

With ac : $P = \frac{V_{rms}^2 R}{Z^2} \Rightarrow Z^2 = \frac{(10)^2 \times 5}{10} = 50 \Rightarrow Z = 5\sqrt{2}\ \Omega$

Also $Z^2 = R^2 + 4\pi^2 v^2 L^2 \Rightarrow 50 = (5)^2 + 4(3.14)^2 v^2 (10 \times 10^{-3})^2$

$\Rightarrow v = 80Hz.$

Final Answer : (3)

10) **Explaining :** To find the value of quality factor.
Concept : This question is based on sharpness of resonance
Solution :
$$Q = \frac{f_r}{\Delta f} = \frac{6}{9-4} = \frac{6}{5} = 1.2$$
Final Answer : (4)

11) **Explaining :** Which of the following have zero average value in a plane electromagnetic wave? **Concept :** This question is based on
$< E > = 0$
$< B > = 0$
Solution : $E = E_0 \sin(kx - \omega t)$
$\langle E \rangle = 0$
$B = B_0 \sin(kx - \omega t)$
$\langle B \rangle = 0$
Final Answer : (3)

12) **Explaining :** To match the entries of column-I with column-II
Concept : This question is based on AC circuits
Solution :
for graph A : phase diff. between current and voltage is zero.
So circuit can be LCR under resonance
for graph B : I is ahead of V by $\frac{\pi}{2}$. So power factor is zero.
for graph C : I is ahead of V but phase diff. $\neq \frac{\pi}{2}$.
So, it may be LCR or RC.
for graph D : V is ahead of I by $\frac{\pi}{2}$. So power factor is zero.
Final Answer : (1)

13) **Explaining :** To find impedance of circuit.
Concept : This question is based on resonance
Solution :
When C is removed
$$\phi = \frac{\pi}{4}$$
$$\frac{X_L}{R} = \tan\phi = 1$$
$X_L = R$
When L is removed
$$\frac{X_C}{R} = \tan\phi = \tan\frac{\pi}{4} = 1$$
$X_C = R$
$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$
$Z = R = 100\Omega$
Final Answer : (2)

14) **Explaining :** To find the reading of the voltmeter V_3 and ammeter A are respectively
Concept : This question is based on resonance

Solution :

V_3 = 220 volt, $I = \frac{220}{100}$ = 2.2A

Final Answer : (3)

15)

Explain Question:

Find the valve of inductance when voltmeter shows100 volt.

Concept:

Use resonance condition of series LCR circuit.

Solution:

Form circuit $V_R = V_{app.}$ = 100 volt

This is the condition of resonance So $X_L = X_C$

$f = \frac{1}{2\pi\sqrt{LC}}$, given f = 500 Hz, C = 10×10^{-6}F

$$L = \frac{1}{4\pi^2 f^2 C} = \frac{1}{4\pi^2 \times 500 \times 500 \times 10 \times 10^{-6}} = 0.01\,H$$

Final answer: Option (4)

16) **Explaining :** To find the value of inductance

Concept : This question is based on resonance

Solution :

$\omega_r = \frac{1}{\sqrt{LC}}$

$\therefore L \propto \frac{1}{C}$

Final Answer : (3)

17) **Explaining :** To find the dc current

Concept : This question is based on R.M.S. value of current.

Solution :

$I_{rms} = \frac{1.41}{\sqrt{2}} = 1A$

$I_{rms}^2 R(\Delta t) = I^2 R(\Delta t)$

$I = 1A$

Final Answer : (4)

18) **EXPLANATION :** The general equation for a plane electromagnetic wave travelling in the +x direction with the electric field polarized along the y-direction is: $E_y = E_0 \sin\omega\left(t - \frac{x}{c}\right)$

CONCEPT : The equation for a plane electromagnetic wave travelling in the +x direction is $E_y = E_0 \sin\omega\left(t - \frac{x}{c}\right)$, where E^0 is the amplitude and ω is the angular frequency. The angular frequency is $\omega = \frac{2\pi c}{\lambda}$.

Solution :

$$\omega = 2\pi\upsilon = \frac{2\pi c}{\lambda} = \frac{2\pi \times 3 \times 10^8}{6 \times 10^{-3}} = \pi \times 10^{11} \text{rad/s}$$

The equation for the electric field, along y-axis in the electromagnetic wave is given by

$$E_y = E_0 \sin\omega\left(t - \frac{x}{c}\right) = 33\sin\left[\pi \times 10^{11}\left(t - \frac{x}{c}\right)\right]$$

Final Answer : (2)

19) **EXPLANATION :** The speed of light (v) in a medium is related to the speed of light in vacuum (c) and the refractive index (n) of the medium by the formula $v = \frac{c}{n}$. For a non-magnetic medium (where relative permeability $\mu_r = 1$), the refractive index (n) is related to the relative permittivity ($\in_r$) by:

CONCEPT : The speed of light (v) in a medium is $v = \frac{c}{n}$, where n is the refractive index. For an electromagnetic wave, the refractive index of a medium is related to its relative permeability (μ_r) and relative permittivity ($\in_r$) by $n = \sqrt{\mu_r \in_r}$.

Solution :

$$v_2 = \frac{c}{n_2} = \frac{c}{\sqrt{\mu_{r_2} \varepsilon_{r_2}}}$$

$$v_2 = \frac{c}{\sqrt{1 \times 9}} = \frac{c}{3} = 1.0 \times 10^8 \text{ms}^{-1}$$

Final Answer : (1)

20) **Explaining the Question :** To check wavelength of various radiation of emw

Concept : This question is based on Electro-magnetic spectrum.

Solution : γ rays - 10^{-12} in

UV - 10^{-8} m

IR – 10^{-5}m

Radio – 10^3 m

Final answer : (2)

21) **Explaining :** An electromagnetic wave going through vacuum is described by E = E sin(kx – ωt). Which of the following is/are independent of the wavelength?

Concept : This question is based on $k = \frac{2\pi}{\lambda}$.

Solution : $k = \frac{2\pi}{\lambda}$

$$\omega = 2\pi f = \frac{2\pi c}{\lambda}$$

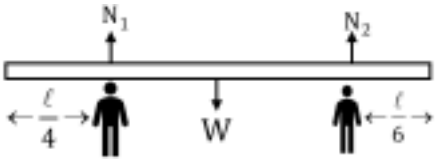
$$\frac{k}{\omega} = \frac{1}{c}$$

Final Answer : (3)

22) **Explaining :** To find ratio of normal reaction at their heads.

Concept : This question is based on concept of torque.

Solution :



τ_{net} = 0 about C.M. of the rod.
torque about W
$W(0) + N_1\frac{\ell}{4} - N_2\frac{\ell}{3} = 0$
$N_1 : N_2 = 4 : 3$
Final Answer : (1)

23) **Explaining :** To find the duration till which angular acceleration must be applied.
Concept : This question is based on concept of dynamics of rotational motion.
Solution : Given moment of inertia 'I' = 1.5 kgm^2
Angular Acc. "α" = 20 Rad/s^2
$= \frac{1}{2} I\omega^2$
$1200 = \frac{1}{2}1.5 \times \omega^2$
$\omega^2 = \frac{1200 \times 2}{1.5} = 1600$
$\omega = 40\ rad/s^2$
$\omega = \omega_0 + \alpha t$
40 = 0 + 20 t t = 2 sec.
Final Answer : (1)

24)

Explain Question: evaluate an assertion and a reason related to a ladder's stability. The assertion claims a ladder is more likely to slip when a person is high up, and the reason attributes this to a larger torque at a higher position.
Concept: This question is based on static equilibrium,
Formula : Torque (τ) = Force (F) × Perpendicular distance (r) from the pivot point.
Net Force = 0
Net Torque (τ_{net}) = 0
Solution: When a person climbs a ladder, their weight creates a downward force. As the person climbs higher, this distance increases, causing the torque to become larger. A greater torque must be balanced by a larger static friction force at the base Since there's a limit to the maximum static friction, the ladder is more likely to slip when the person is high up.
Final answer: (1)

25) **Explaining :** To find the magnitude of this torque.
Concept : This question is based on concept of torque.
Solution : $\tau = \frac{dL}{dt} = \frac{4A_0 - A_0}{4} = \left(\frac{3A_0}{4}\right)$
Final Answer : (3)

26) **Question Explanation :** We must find how many times the radius of gyration increases

when a solid disc is reshaped into a ring of radius 2R

Concept : Radius of gyration

Solution:
Moment of inertia of the disc of mass M and radius R

$$I_{disc} = Mk_1^2 = \frac{1}{2}MR^2$$

$$\Rightarrow \quad k_1 = \frac{R}{\sqrt{2}}$$

And for the ring of same mass but radius 2R,

$$I_{ring} = Mk_2^2 = M(2R)^2$$

$$\Rightarrow \quad k_2 = 2R$$

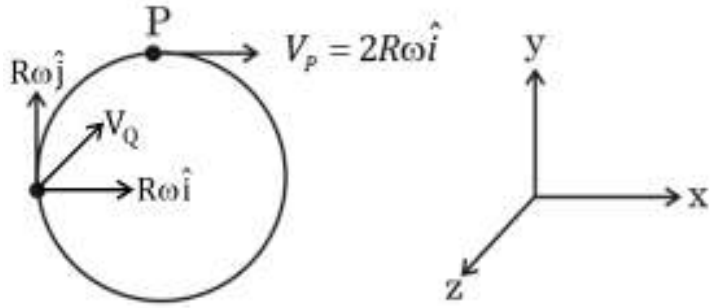
Thus, $\frac{k_2}{k_1} = 2\sqrt{2}$

Final Answer: Option(2)

27) **Question Explanation:**
The problem asks to find the ratio of speeds at point P and point Q for a ball rolling on the ground without slipping.

Concept:
This question is based on concept of instantaneous centre of rotation.

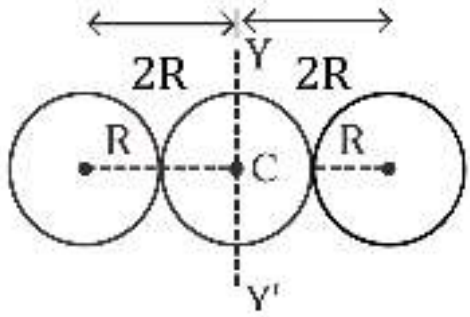
Solution:



$$V_Q = R\omega\hat{j} + R\omega\hat{i} \quad \because \frac{|V_P|}{|V_Q|} = \frac{2}{\sqrt{2}} = \frac{\sqrt{2}}{1}$$

Final Answer:
The correct option is (1)

28) **Explain Question:** find the total moment of inertia (M.I.) of a system of three identical spherical shells. The shells are arranged in a row, and we need to calculate the total M.I. about the vertical axis YY′ which passes through the center of the middle shell.
Concept: This question is based on Moment of Inertia- Parallel Axis Theorem
Formula : $I = I_{CM} + Md^2$
Solution:



Using parallel axis theorem

$$I = \frac{2}{3}MR^2 + 2\left[\frac{2}{3}MR^2 + M(2R)^2\right]$$

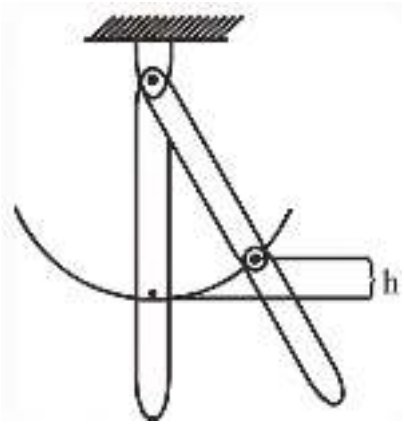
$I = 10MR^2$

Final answer :- (4)

29) **Explaining :** To find upto what height centre of mass will rise.

Concept : This question is based on concept of centre of mass.

Solution :



$$\frac{1}{2}I\omega^2 = mgh$$

$$\frac{1}{2}\frac{m\ell^2}{3}\omega^2 = mgh$$

$$h = \frac{1}{6}\frac{\ell^2\omega^2}{g}$$

Final Answer : (3)

30) **Explaining :** To find the total angle rotated by the wheel (in radian).

Concept : This question is based on concept of kinematics of rotational motion.

Solution :

For acceleration $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2} \times 2 \times (20)^2 = 400$ Rad.

$\omega = 0 + \alpha t = 2 \times 20 = 40$ Rad/sec

For uniform motion,

$\theta = \omega \times t = 40 \times 10 = 400$ Rad.

For retardation

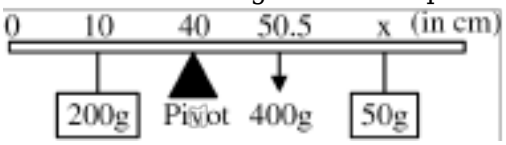
$$\theta = \frac{1}{2}t(\omega_i + \omega_f) = \frac{1}{2} \times 20\ (40 + 0) = 400\ \text{Rad.}$$

Total angle of rotation = 1200 Rad.

Final Answer : (2)

31) **Explaining :** To find the power associated with torque due to the force.
Concept : This question is based on concept of power.
Solution : $\vec{\tau} = \vec{r} \times \vec{f}$ & $P = \vec{\tau}.\vec{\omega}$ $\Rightarrow P = (\vec{r} \times \vec{f}).\vec{\omega}$
Final Answer : (1)

32) **Explaining :** To find the position, where a 50g mass must be suspended so as to balance the scale.
Concept : This question is based on concept of meter scale.
Solution : Let 50 g mass is suspended at x cm mark

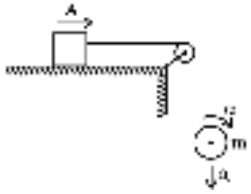


From principle of moments :
(40 – 10)200 g = (50.5 – 40) × 400g + (x – 40)50g
6000 = 4200 + 50x – 2000
⇒ 50x = 3800 ⇒ x = 76 cm
Final Answer : (3)

33) **Explain Question:**
To find the relation between a_c & A .

Concept:
Condition of no slipping.

Solution:



$a_c - r\alpha = A$
$\Rightarrow a_c = A + r\alpha$

Final Answer: option(4)

34) **Explain Question :**
To find out MOI of a semi-circular ring of radius 'R', made of wire of length 'L' and mass 'm'.

Concept:
Moment of Inertia

Solution:
MI of given ring = $\frac{MR^2}{2}$

Where $\pi R = L \Rightarrow R = \frac{L}{\pi}$

So, $I = \frac{ML^2}{2\pi^2}$

Final answer: Option (1)

35) **Explaining :** Calculate the power required by an engine to maintain a rotor at angular speed 200 rad/s with torque 180 Nm.

Concept : This question is based on Power $P = \tau\omega$

Solution :

Here, $\omega = 200$ rad s^{-1} $\tau = 180$ N-m

Using the relation, $P = \tau\omega$, we get

$P = 180 \times 200$

$= 36000$ W $= 36$ kW

Final Answer : (3)

36)

Explain Question: A spherical shell rolls without slipping on a rough inclined plane in one case and slides down a smooth inclined plane of the same inclination in another. find the ratio of its linear accelerations in these two cases.

Concept: pure rolling dynamics on an incline plane

Formula :

$$a_r = \frac{g\sin\theta}{1 + k^2/R^2}$$

$a_{sliding} = g\sin\theta$

Solution:

$$a_r = \frac{g\sin\theta}{1 + K^2/R^2}$$

For spherical shell, $\frac{K^2}{R^2} = \frac{2}{3}$

$a_r = \frac{3}{5} g \sin\theta$ ------- (1)

but $a_S = g \sin\theta$ -------- (2)

dividing the two equations

$$\frac{a_r}{a_S} = \frac{3}{5}$$

Final answer: (2)

37)

Explaining the Question :- What happens to a body when a torque acts on it.

Concept:- This question is based on rotational dynamics

Formula :- $\tau = I\alpha$
Solution :- According to the formula $\tau = I\alpha$, if a torque is acting on a body ($\tau = 0$), and the body has a moment of inertia ($I = 0$), it will experience an angular acceleration ($\alpha = \tau/I$). Angular acceleration is the rate of change of angular velocity. A body undergoing angular acceleration is either speeding up or slowing down its rotation. This means its state of rotational motion is changing. Therefore, the body gets angular acceleration.
The final answer is that when torque is acting on a body, the body gets angular acceleration
Final Answer :- 4

38) **Explaining :** To find the initial angular acceleration of the system.
Concept : This question is based on concept of torque.
Solution : Torque about the point 'O' will be
$\tau_{ext} = mg \times R + mg \times 3R = 4mgR$
So the initial angular acceleration of system will be

$$\alpha = \frac{\tau_{ext}}{1} = \frac{4mgR}{\frac{34mR^2}{3}} = \frac{6g}{17R}$$

[Here $I = \frac{m(2R)^2}{3} + \left(mR^2 + m(3R)^2\right)$

$$= \frac{4mR^2 + 3mR^2 + 27mR^2}{3} = \frac{34mR^2}{3}$$

Final Answer : (1)

39) **Explaining :** To find the moment of inertia of the remaining portion of the disc.
Concept : This question is based on concept of **Moment of Inertia of a Solid Disc.**
The moment of inertia of a solid disc of radius R and mass M about its center is:

$$I_{original} = \frac{1}{2}MR^2$$

Solution : $I_{removed} = \frac{1}{2}\left(\frac{M}{4}\right)\left(\frac{R}{2}\right)^2 + \left(\frac{M}{4}\right)\left(\frac{R}{2}\right)^2 = \frac{3}{32}MR^2$

$$I_{total} = \frac{1}{2}MR^2$$

$$I_{remaining} = I_{total} - I_{removed}$$

$$= \frac{MR^2}{2} - \frac{3MR^2}{32} = \frac{13MR^2}{32}$$

Final Answer : (1)

40) **Explaining :** To find the moment of inertia of rods about an axis passing through O.
Concept : This question is based on concept of perpendicular axis theorem.
Solution :

- Moment of inertia about an axis passing through an end = $\frac{ML^2}{3}$
- $I_{given\ system} = \frac{ML^2}{3} + \frac{ML^2}{3} = \frac{2ML^2}{3}$

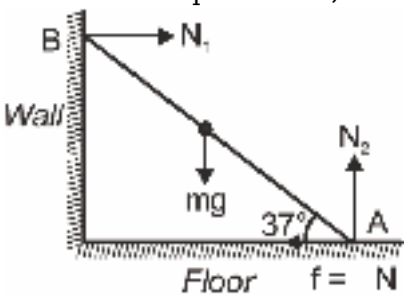
Final Answer : (2)

41) **Explaining :** To find the fraction of total energy associated with its rotation.
Concept : This question is based on concept of Total kinetic energy of rolling body.

Solution :
Both assertion and reason are true and reason is the correct explanation of assertion.

$$\text{Total KE} = \frac{1}{2}m\nu^2 + \frac{1}{2}I\omega^2$$
$$= \frac{1}{2}m\nu^2 + \frac{1}{2}mK^2\omega^2$$
$$= \frac{1}{2}m\omega^2(R^2 + K^2)$$
$$\text{KE of rotation} = \frac{1}{2}I\omega^2\frac{1}{2}(mK^2)\omega^2$$
$$\therefore \quad \frac{\text{KE of rotation}}{\text{Total KE}} = \frac{K^2}{R^2 + K^2}$$ **Final Answer : (1)**

42) **Explaining :** To find minimum friction co-efficient between floor and ladder for the given equilibrium.
Concept : This question is based on concept of rotational equilibrium.
Solution :
For vertical equilibrium ;

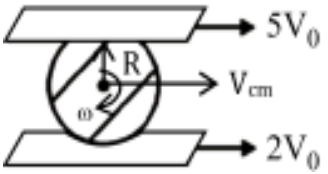


$mg = N_2$ .....(i)
For horizontal equilibrium ;
$\mu N_2 = N_1$ .....(ii)
Torque about A ;
$N_1 \times 5 \sin 37° = mg \times 2.5 \cos 37°$ .....(iii)
Solving ; $\mu = 2/3$
Final Answer : (4)

43) **Question Explanation:** Find the angular velocity ω of a cylinder that is rolling given that the velocity at its top & bottom are $5V_0$ & $2V_0$ respectively.

Concept: Pure rolling on moving surface

Solution:



Velocity at a point on rotating body $= V_{Trans} + V_{Rot}$
At top, $V_{cm} + R\omega = 5V_0$ ...(1)
At bottom, $V_{cm} - R\omega = 2V_0$ ...(2)

Equation (1) + (2)

$2R\omega = 3V_0$

$$\omega = \frac{3V_0}{2R}$$

Final Answer: Option (1)

44) **Explain Question :**
Calculating the ratio of angular momentum for two bodies.

Concept:
Relation between Kinetic Energy and Angular Momentum.

Solution:
Rotational kinetic energy (K) is related to angular momentum (L) and moment of inertia (I). If the kinetic energies are the same, the ratio of their angular momenta depends on the square root of the ratio of their moments of inertia.
Calculation:
Formula: $K = \frac{L^2}{2I} \Rightarrow L = \sqrt{2IK}$
Given: $K_1 = K_2, I_1 = 9, I_2 = 1$

$$\frac{L_1}{L_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{9}{1}} = \frac{3}{1}$$

Final Answer: Option (1).

45) **Explaining :** To find kinetic energy of the system.
Concept : This question is based on concept of moment of inertia.
Solution :

$$KE = \frac{L^2}{2I}$$

$$I \rightarrow 2I$$

$$KE \rightarrow \frac{KE}{2}$$

Final Answer : (4)

CHEMISTRY

46) **Explaining :** In inert pair effect, which pair of electrons are said to be inert:
Concept : This question is based on inert pair effect.
Solution :
ns^2 electrons don't participate in bonding due to poor shielding effect of filled (n-2)f and (n-1)d orbitals. While their poor shielding causes the inert pair effect, these electrons themselves are not the ones considered the inert pair.
Final Answer : (4)

47) **Explaining :** The ion that cannot undergo disproportionation is
Concept : This question is based on For an element to undergo disproportionation reaction, it

must be in an intermediate oxidation state.
Solution :
In the oxoanions of chlorine:

A. ClO^- (+1)

B. ClO_2^- (+3)

C. ClO_3^- (+5)

The chlorine exists in states that allow it to both increase and decrease its oxidation state, allowing disproportionation.
However, in ClO_4^-, the oxidation state of chlorine is **+7**. Because +7 is the **maximum possible oxidation state** for chlorine, it can only be reduced, not oxidized further. Therefore, it cannot undergo a disproportionation reaction.
Final Answer : (1)

48) **Explaining :** Evaluate both the statements.
Concept : This question is based on bond angle.
Solution :
Bond angle ∝ electronegativity of C.A.
Down the group 15, metallic character increases.
Final Answer : (2)

49) **Explaining :** Sulphur dioxide may be recognized by its?
Concept : This question is based on sulphur dioxide (SO_2) is its behavior as a strong reducing agent and its physical volatility
Solution :
Sulphur dioxide may be recognized by its
(1) Characteristic pungent smell of burning sulphur
(2) Ability to turn dichromate paper green
(3) Ability to decolourize acidified $KMnO_4$ solution to colorless
Final Answer : (4)

50) **Explaining :** Which of the following is not correctly matched?
Concept : This question is based on nature of oxide .
Solution :
Generally, as you move down a group in the periodic table, the metallic character of elements increases. Consequently, the acidic nature of their oxides decreases, and the basic character increases. Elements in the middle of these groups (like (Ga), (Al), (Sn), (Pb)) often exhibit dual behavior and form **amphoteric** oxides
Final Answer : (4)

51) **Explaining :** The IUPAC name of complex $K_3[Al(C_2O_4)_3]$ is;
Concept : This question is based on IUPAC naming.

Solution : Correct IUPAC name for the compound is potassium trisoxalatoaluminate (III)
Final Answer : (4)

52) **Explaining :** Of the following, which ligand does not posses the name suggested by IUPAC when it acts as a ligand in complex?
Concept : This question is based on rule of IUPAC naming.
Solution : NH_3 - Ammine.
Final Answer : (2)

53) **Explaining :** The compound which does not show paramagnetism is;
Concept : This question is based on magnetic nature of compounds.
Solution : Ag^+ :- $4d^{10}$, no unpaired electrons are present in $[Ag(NH_3)_2]Cl$.
Final Answer : (2)

54) **Explaining :** What are the geometric shape and the oxidation number of the copper atom, respectively, for the complex ion,
$[Cu(NH_3)_4(H_2O)_2]^{+2}$?
Concept : This question is based on geometry, oxidation number.
Solution :
$[Cu(NH_3)_4(H_2O)_2]^{+2}$
Cu^{+2} : $3d^9$
CN = 6
Geo = Octahedral
Final Answer : (4)

55) **Explaining :** The solution of the complex $[Cu(NH_3)_4]SO_4$ in water:
Concept : This question is based on addition salts.
Solution :
$[Cu(NH_3)_4]SO_4$ is a coordinate salt.
It will not dissociate in water completely into its constituent ion. So, it will give the tests of SO_4^{2-} ion only.
Final Answer : (3)

56) **Explaining :** How many of following metals impart a characteristic colour to the Bunsen flame?
Concept : This question is based on Flame test.
Solution :
Both beryllium and magnesium have exceptionally small atomic sizes and high ionization enthalpies. Their valence electrons are very tightly held by the nucleus, meaning the relatively low thermal energy of a standard Bunsen burner flame is insufficient to excite them. Consequently, they do not emit light in the visible spectrum
Final Answer : (4)

57) **Explaining :** A is almost insoluble in concentrated ammonia solution. identify the anion in the salt.
Concept : This question is based on Qualitative analysis.

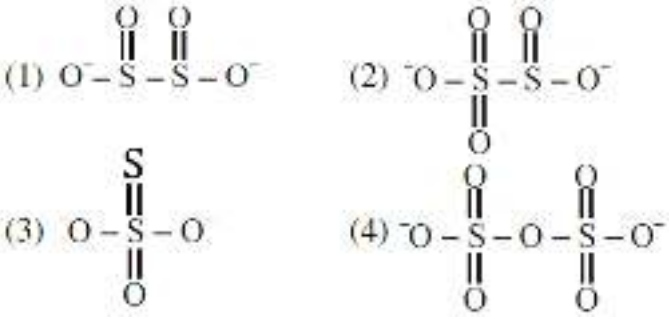
Solution :
Yellow Precipitate (A): When the salt reacts with silver nitrate ($AgNO_3$), it forms a bright yellow precipitate of silver iodide (AgI).
Solubility: Unlike other yellow precipitates (such as silver phosphate or silver chromate), (AgI) is **almost insoluble** in concentrated ammonia solution because its crystal lattice energy is exceptionally high.
Final Answer : (3)

58) **Explaining :** Experiment of detection of S^{-2} is given below in alkaline medium
$S^{2-} + [Fe(CN)_5NO]^{2-} \rightarrow [Fe(CN)_5NOS]^{4-}$
Select the correct statement about the experiment.
Concept : This question is based on Confirmatory Test for Sulphide ions (S^{2-})
Solution : The newly formed complex $[Fe(CN)_5NOS]^{4-}$ colour is **purple or violet.**
Final Answer : (1)

59) **Explaining :** Which one of the following reactions does not occur ?
Concept : This question is based on oxidizing power of halogens.
Solution :
F^- cannot be oxidised by Cl_2
(F_2 is stronger O.A. than Cl_2)
Final Answer : (2)

60) **Explaining :** Which of the following is not peroxide.
Concept : This question is based on peroxide is a compound that contains an oxygen-oxygen single bond.
Solution : PbO_2 is not a peroxide. PbO_2 is a strong oxidising agent.
Final Answer : (3)

61) **Explaining :** There is no S—S linkage in
Concept : This question is based on structure of oxyacids.
Solution : -



Final Answer : (4)

62) **Explaining :** Which of the following statements are correct for complex $K_2[Zn(CN)_2(O_2)]$?
Concept : This question is based on IUPAC nomenclature, isomerism, and valence bond theory of coordination compounds.
Solution :
(i) Potassium dicyanidodi(superoxido)zincate(II).
(ii) Tetrahedral complexes do not exhibit geometrical isomerism.

(v) Because the inner (3d) orbitals are fully occupied and unavailable for hybridization, it uses outer (4s) and (4p) orbitals (sp^3) hybridization. This makes it an **outer orbital complex**.
Final Answer : (1)

63) **Explaining :** Find the correct statements.
Concept : This question is based on properties of d-block metals.
Solution :
(i) Scandium has a stable (+3) oxidation state and forms (Sc_2O_3) instead.
(ii) As the oxidation state of the metal increases, its polarizing power increases, resulting in greater covalency and a shift in character from basic to amphoteric or acidic.
(iii) CrO is basic, but (Cr_2O_3) is amphoteric (not acidic)
(iv) In an acidic medium, Mn(VI) species (such as manganate, (MnO_4^{2-}) are highly unstable and readily undergo disproportionation to form (Mn(VII)) (permanganate, (MnO_4^-) and (Mn(IV)) (manganese dioxide, (MnO_2).
Final Answer : (4)

64) **Explaining :** Evaluate both the statements.
Concept : This question is based on isomerism.
Solution : Tetrahedral complexes do not show geometrical isomerism because the relative position of unidentate ligands attach to central metal atom are same w.r.t. each other.
Final Answer : (1)

65) **Explaining :** Find out x – y + 1
Concept : This question is based on properties of d-block elements.
Solution :
(1) V have max ΔH (Correct): Vanadium (V) has the maximum enthalpy of atomization (Δ_a H) in the 3d series.
(2) Ti have min ΔH (Incorrect): Titanium does not have the minimum ionization enthalpy.
(3) Sc have max metallic radii (Correct): Scandium has the largest atomic/metallic radii in the 3d series because effective nuclear charge increases and shielding decreases across a period.
(4) E° value for M^{+2}/M is negative for V and Cr (Correct): Standard reduction potentials (E°) for M^{2+}|M are negative across almost the entire 3d series (including V and Cr), except for copper.
(5) Density of Cu ≃ density of Ni (Correct): Density of Cu = density of Ni = 8.9 g/ml
Calculation:
Number of correct properties (x) = (4)
Number of incorrect properties (y) = (1)
x – y + 1 = 4 – 1 + 1 = 4
Final Answer : (3)

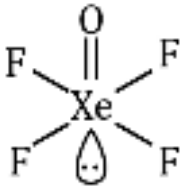
66) **Explaining :** Lanthanide contraction means:
Concept : This question is based on periodic trends in the f-block elements.
Solution :
Lanthanide contraction means the steady **decrease in atomic size** of elements **after lanthanum**.

A. **Poor Shielding:** Inner **(4f) electrons** shield nuclear charge poorly.

B. **Stronger Pull:** The nucleus pulls outer electrons **closer**.

Final Answer : (1)

67) **Explaining :** The geometry of $XeOF_4$ by VSEPR theory is:
Concept : This question is based on VSEPR theory.
Solution :



Final Answer : (1)

68) **Explaining :** Analyse both the statements
Concept : This question is based on type of bonding
Solution :
The sum of the first four ionization enthalpies of carbon is very high, So it can't easily loose electrons therefore, compounds in +4 oxidation state are generally covalent in nature.
Final Answer : (1)

69) **Explaining :** In which of the following reactions the hybridisation state of underlined atom is same in reactants and products?
Concept : This question is based on hybridisation.
Solution :

$$\underset{sp^3}{\underline{N}H_3} + H^+ \rightarrow \underset{sp^3}{NH_4^+}$$

Hybridisation of N is same in reactant and product.
Final Answer : (1)

70) **Explaining :** Which of the following is a bidentate ligand?
Concept : This question is based on denticity of the ligand.
Solution : Ethylenediamine is a bidentate ligand.
Final Answer : (1)

71) **Explaining :** Which of the following is correct for the complex $[NiBr_2(PPh_3)_2]$?
Concept : This question is based on geometry of $[NiBr_2(PPh_3)_2]$
Solution :
$[NiBr_2(PPh_3)_2]$
Ni^{2+} : $3d^8$
CN = 4
Geo. = tetrahedral (WFL)
unpaired e^- = 2
Final Answer : (1)

72) **Explaining :** Bucky ball or Buckminster fullerene is:
Concept : This question is based on allotrope of carbon.
Solution : All of these statements are true for fullerene.

Final Answer : (4)

73) **Explaining :** In the coordination compound $K_4[Ni(CN)_4]$, the oxidation state of nickel is:
Concept : This question is based on oxidation state.
Solution : $[Ni(CN)_4]^{4-}$: Ni^0
Final Answer : (2)

74) **Explaining :** The following statements are TRUE about crystal field theory except:
Concept : This question is based on CFT
Solution :
Statements (1), (2), and (3) are true limitations of CFT. CFT treats metal-ligand interactions as strictly electrostatic (ionic) and considers only the central metal's (d)-orbitals. Consequently, it fails to explain (pi)-bonding (or back-bonding), the partial covalent nature of the metal-ligand bond, and the position of some neutral ligands in the spectrochemical series (e.g., why (H_2O) is a stronger ligand than (OH^-))
Statement (4) is false. CFT primarily focuses only on the (d)-orbitals of the central metal ion. The valence (s) and (p) orbitals of the metal are *not* considered during CFT bond formation and orbital splitting calculations.
Final Answer : (4)

75) **Explaining :** In SiF_6^{2-} and $SiCl_6^{2-}$, which one is known and why?
Concept : This question is based on Steric hindrance & Coordination number.
Solution :
SiF_6^{2-} exists and is known, while $SiCl_6^{2-}$ does not. This is because fluorine is smaller, allowing six atoms to easily fit around the central silicon atom, whereas large chlorine atoms cause too much steric hindrance.
Final Answer : (1)

76) **Explaining :** Amongst the following the most stable compound is:
Concept : This question is based on stability of complexes.
Solution : Stability of complex ∝ No. of chelating rings.
Final Answer : (2)

77) **Explaining :** Which of the following is an organometallic compound:
Concept : This question is based on organometallic compound.
Solution :
Zeise salt is organometallic compound.
$[PtCl_4(C_2H_4)]$ is π-bonded OMC.
Final Answer : (2)

78) **Explaining :** For the confirmatory test of bromide ion, we acidify 1 mL of sodium carbonate extract with dil. HNO_3 (or take 1 mL water extract) and add silver nitrate solution. A pale yellow precipitate is obtained of-
Concept : This question is based on confirmatory test of bromide.
Solution :
When a solution containing bromide ions (Br^-) is acidified with dilute nitric acid (HNO_3) and

treated with silver nitrate solution ($AgNO_3$), a precipitation reaction takes place. The silver ions (Ag^+) react with the bromide ions to form **silver bromide (AgBr)**, which separates out as a characteristic **pale yellow precipitate**
Final Answer : (2)

79) **Explaining :** Evaluate both the statements.
Concept : This question is based on preparation of Mohr's salt.
Solution :
Dilute H_2SO_4 is preferred because sulfate ions are already part of the salt. Chloride ions may interfere in later reactions and titrations.
Assertion is false but Reason is also false because chloride ions may interfere.
Final Answer : (4)

80) **Explaining :** Find the incorrect match:
Concept : This question is based on dimerization
Solution : Hyb. of I in I_2 Cl_6 is sp^3d^2 with 2 lp that's why square planar geometry
Final Answer : (3)

81) **Explaining :** Among Fe^{3+}, Pb^{2+}, Cu^{2+} and Ni^{2+} identify the one that gets precipitated out while passing H_2S in presence of NH_4OH as group reagent. The highest possible oxidation state of the corresponding metal is
Concept : This question is based on qualitative analysis.
Solution :
While passing H_2S in the presence of NH_4OH, the medium becomes basic.
In basic medium, the concentration of S^{2-} ions increases, so sulphides of some metal ions get precipitated.
Now let us check the given ions:
Fe^{3+} : does not precipitate here as sulphide in this group
Pb^{2+} : belongs to earlier group and is precipitated as chloride, not here
Cu^{2+} : precipitates as CuS
Mn^{2+} : can precipitate as Ni in ammoniacal medium
But in classical qualitative analysis, passing H_2S in presence of NH_4OH is used for precipitating group IV cations like Mn^{2+}, Zn^{2+}, Ni^{2+}, Co^{2+} as sulphides.
So the ion identified here is Ni^{2+}.
The corresponding metal is Nickel, and its highest oxidation state is: +4
Final Answer : (2)

82) **Explaining :** Which of the following complex formed by Cu^{+2} ions is most stable?
Concept : This question is based on stability of complexes.
Solution : Stability of complex ∝ Synergic bonding.
Final Answer : (2)

83) **Explaining :** The compounds $[Co(SO_4)(NH_3)_5]Br$ and $[Co(SO_4)(NH_3)_5]Cl$ represent :-
Concept : This question is based on isomerism.
Solution : $[Co(SO_4)(NH_3)_5]Br$ and $[Co(SO_4)(NH_3)_5]Cl$ represent no isomerism as their molecular formula are different.

Final Answer : (4)

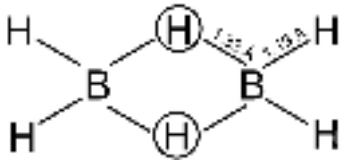
84) **Explaining :** Among the following transition metal ions, the one where all metal ions have 4 unpaired electrons:
Concept : This question is based on electronic filling of electrons of d-block metal ions.
Solution :
Cr^{+2} : d^4
Mn^{+3} : d^4
Fe^{+2} : d^6
Final Answer : (2)

85) **Explaining :** Analyse both the statements about $K_2Cr_2O_7$.
Concept : This question is based on characteristics of a primary standard.
Solution :
Statement I is True: Potassium dichromate ($K_2Cr_2O_7$) is highly stable, non-hygroscopic, and can be weighed accurately. Sodium dichromate ($Na_2Cr_2O_7$) is hygroscopic (absorbs moisture) and therefore cannot be accurately weighed to serve as a primary standard.
Statement II is False: Sodium dichromate is significantly more soluble in water than potassium dichromate. However, this solubility difference is not the reason $K_2Cr_2O_7$ is chosen as a standard.
Final Answer : (4)

86) **Explaining :** When excess of KCN is added into $CuSO_4$ solution then product obtained is
Concept : This question is based on complex formation
Solution : $KCN + CuSO_4 \rightarrow K_3[Cu(CN)_4]$
Final Answer : (4)

87) **Explaining :** Which of the following pairs of isomers and types of isomerism are correctly matched?
Concept : This question is based on isomerism.
Solution : All statements are correctly matched with their respective forms of structural isomerism.
Final Answer : (4)

88) **Explaining :** The molecular shapes of diborane is shown : Consider the following statements for diborane :
Concept : This question is based on diborane structure.
Solution :



Final Answer : (1)

89) **Explaining :** Which one of the following complexes has the largest C. F. S. E. (crystal field stabilization energy)?

Concept : This question is based on C.F.S.E.
Solution :

Complex Ion	Metal Ion	*d*-Electron Count	High-Spin Configuration	CFSE Calculation	Total CFSE
(1) $[Mn(H_2O)_6]^{2+}$	Mn^{2+}	$3d^5$	$t_{2g}^3 e_g^2$	$[3\times(-0.4)+2\times(+0.6)]\Delta_o$	$0\ \Delta_o$
(2) $[Cr(H_2O)_6]^{2+}$	Cr^{2+}	$3d^4$	$t_{2g}^3 e_g^1$	$[3\times(-0.4)+1\times(+0.6)]\Delta_o$	$-0.6\ \Delta_o$
(3) $[Mn(H_2O)_6]^{3+}$	Mn^{3+}	$3d^4$	$t_{2g}^3 e_g^1$	$[3\times(-0.4)+1\times(+0.6)]\Delta_o$	$-0.6\ \Delta_o$
(4) $[Cr(H_2O)_6]^{3+}$	Cr^{3+}	$3d^3$	$t_{2g}^3 e_g^0$	$[3\times(-0.4)+0\times(+0.6)]\Delta_o$	$-1.2\ \Delta_o$

Final Answer : (4)

90) **Explaining :** Which of the following is not a bidentate ligand?
Concept : This question is based on type of ligand.
Solution : NH_2–NH_2 is monodentate ligand.
Final Answer : (2)

BIOLOGY

91) **Explaining the Question:** This question asks which statement correctly describes the activation of Bt-toxin produced by *Bacillus thuringiensis*.
Concept: Bt Cotton and Bt Toxin (Microbial Bio-Control Agents).
Solution : Correct. - Bt-protoxin becomes an active toxin only in the alkaline pH of the insect midgut, where it gets solubilised and cleaved.
Final Answer: (3)

92) **Explanation :** The question asks to fill in the blanks in four statements related to biotechnology and pest resistance in plants.
Concept : The concept based on Application of Biotechnology in agriculture.
Final Answer : (2)

93)

Question Explanation: We must evaluate each statement about rDNA technology and Bt toxins, then select the correct option.
Concept: Application of Biotechnology in agriculture.
Solution:

A. rDNA technology helps improve crops by increasing yield, nutrition, and disease resistance - TRUE

B. Bt cotton protects against bollworm due to Bt toxin production - TRUE

C. *Bacillus thuringiensis* produces Cry toxin as protoxin, so it does not harm the bacterium - TRUE

D. Protoxin becomes active in **alkaline** pH(MIDGUT), not acidic pH(STOMACH) — FALSE

Final answer : (3)

94) **Solution :** Transgenic host expressing two complementary RNAs that initiated RNAi.
Final Answer : (3)

95) **Explain Question:** The question is asking to identify the correct definition of a transgenic animal.
Concept: Transgenic animals
Solution: A transgenic animal is produced by introducing foreign DNA into the nucleus of a fertilized egg (zygote). Since all cells of the organism develop from this zygote, the foreign gene is present in all cells, including germ cells.
Final answer : (2)

96)

Question Explanation: Identify the term for illegal use of biological resources without permission.

Concept: Bioethics

Solution: The correct answer is option 4

A. Biopiracy means developing or using biological materials (like plants, microbes) from a region without compensating the native people. Hence, it is unlawful exploitation.

Final Answer: Option 4

97) **Explanation :** Identify the wrongly matched pairs in the given table related to transgenic organisms and their characteristics.
Concept : The concept based on Application of Modern biotechnology.
Solution : The wrongly matched pairs are **:**
(ii) Flavr Savr transgenic tobacco with delayed ripening (it's a transgenic tomato)
(iii) Bt-cotton with bacterial and viral resistance (it's resistant to certain insects pests)
(iv) 'Rosie' transgenic cow produced milk contained human alpha Lactoglobulin (it's human alpha Lactalbumin).
Final Answer : (2)

98)

Explaining the Question: This question asks about which Indian government organisation approves and monitors the safety and validity of GM (genetically modified) research and

release of GMOs.

Concept: *GMOs and Biosafety Regulations in India (Biotechnology)*.

Solution:
The Genetic Engineering approval Committee, or GEAC for short, is the Indian government's go-to authority when it comes to overseeing and making decisions about genetically modified (GM) research. They play a crucial role in ensuring the safety of introducing genetically modified organisms (GMOs) into the public sphere. Operating under the Ministry of Environment, Forest and Climate Change (MoEFCC), GEAC carefully assesses environmental risks before giving the green light for GM crops and products.
The correct answer is: 2. GEAC (Genetic Engineering Approval Committee)

Final Answer : Option (2)

99) **Question Explanation:** The question asks which protein was enriched in the milk of the first transgenic cow, *Rosie*.
Concept: The question is based on transgenic animals .
Solution:

A. Rosie was the first transgenic cow.
B. Her milk was enriched with **human alpha-lactalbumin**, a protein beneficial for infant nutrition.

Final Answer : (2)

100)

Question Explanation: Identify the true statement about RNA interference (RNAi).
Concept: Biotechnology

Solution: The correct answer is option 3 because the source of complementary RNA that triggers RNAi can come from RNA viruses or transposons that replicate via RNA intermediates.

A. Option 1 is false because RNAi occurs only in eukaryotes, not all living organisms.

B. Option 2 is false because RNAi silences mRNA, not DNA due to double stranded RNA.

C. Option 4 is false because nematode-specific genes are introduced using *Agrobacterium* vectors, not via the pBR322 vector.

Final answer: (3)

101)

Question Explanation: The question asks which peptides are present in human insulin when it exists as a prohormone.

Concept: The question is based on human insulin.
Solution: The correct answer is 1. A, B, C Peptide.

A. Human insulin is first synthesized as **proinsulin**, an inactive precursor.
B. Proinsulin contains **A chain, B chain, and a connecting C-peptide**.
C. During maturation, the **C-peptide is removed**, leaving only A and B chains linked by disulphide bonds to form active insulin.

Proinsulin is a single polypeptide chain that folds to form three distinct regions: the A-peptide, the B-peptide, and the C-peptide.

Final Answer : (1)

102) **Explain Question :** Identify which is not a valid application of transgenic animals.
Concept : This question is based on Transgenic animals.
Solution : ELISA - Molecular diagnosis does not involve transgenic animals; it relies on lab techniques like PCR.
Final answer : (4)

103)

Solution ; This technique for molecular diagnosis relies on the interaction between antigens and antibodies.
ELISA, which stands for Enzyme-Linked Immunosorbent Assay, is a popular method in molecular diagnostics. It helps identify the presence of antigens—those pesky proteins from pathogens—or antibodies that our immune system produces in response. You'll often find it being used to diagnose infections such as HIV, COVID-19, and Hepatitis
Final Answer : (3).

104) **Solution :** PCR is routinely used to detect HIV in suspected AIDS patients.
Final Answer : (3)

105) **Solution :** Basmati rice is distinct for its unique aroma and flavour.
The Indian parliament has made 2nd amendment in Indian patents bill. In 1997, *Rosie* was developed. Rosie milk was alpha-Lactalbumin enriched.
Final Answer : (1)

106) **Question Explanation:** Identify which of the given statements about recombinant therapeutics and Basmati rice are correct.
Concept: The question is based on biotechnology applications and biopiracy issues.
Solution:

A. **Statement A** → Correct. Recombinant therapeutics like insulin, growth hormones, interferons, etc. have been approved globally, around 30 in number.
B. **Statement B** → Correct. India has more than 27 documented traditional varieties of Basmati rice.

Both statements are true.

Final answer: Option 1 (Both A and B are true)

107) **Question Explanation :** The question asks which statement correctly describes the permanent cure for ADA (Adenosine Deaminase) deficiency.
Concept : Gene Therapy (ADA deficiency treatment)
Solution :

A. **Statement I:** ADA enzyme is transferred to patient for permanent cure → Incorrect. Enzyme replacement therapy (giving ADA enzyme injections) provides temporary relief, not a permanent cure. ADA gene is transferred.

B. **Statement II:** ADA deficiency can be permanently cured by ADA enzyme replacement therapy and bone marrow transplantation → Incorrect.

    A. Enzyme replacement therapy is not permanent.

    B. Gene therapy at only embryogenic stages is considered permanent cures.

Thus, both statements are wrong.

Final Answer : (4)

108)

Question Explanation: The question asks whether the Assertion and Reason are true, and whether the Reason explains the Assertion.
Concept: Biotechnological Applications In Agriculture, Tissue Culture
Solution:

A. Assertion (A): Biofortification is used to increase the nutrient value of crops. True – Biofortification aims to enhance vitamins, minerals, and protein content through breeding.
B. Reason (R): Meristem culture is used to obtain virus-free plants. True – Meristem tissues are naturally virus-free, so culturing them yields virus-free plants. But R does NOT explain A because obtaining virus-free plants has no connection to increasing nutrient content.

Final answer: Option 2.

109) **Question Explanation:** Match the events in biotechnology with their correct years.
Concept: The question is based on biotechnology and biopiracy.
Solution:
(i) First clinical gene therapy → 1990 (a)
(ii) First made recombinant DNA → 1972 (b)
(iii) Eli-Lilly formed human insulin → 1983 (c)
(iv) American company got patent rights on Basmati rice → 1997 (d)
So the correct matching is: i – a, ii – b, iii – c, iv – d
Final answer: (2)

110) **Question Explanation:** The question asks what *Agrobacterium tumefaciens* is commonly known as.
Concept: The question is based on biotechnology and genetic engineering in plants.
Solution: The correct answer is option 1.

A. *Agrobacterium tumefaciens* is a soil bacterium that causes crown gall disease in plants.
B. It naturally transfers a segment of its Ti plasmid (T-DNA) into the plant genome. Because of this ability, it is called the **natural genetic engineer of plants**.

Final Answer: Option 1

111)

Question Explanation: Identify the enzyme that is incorrectly matched with its function.
Concept: Biotechnology Tools
Solution:

A. Ligase – molecular glue → correct

B. Endonuclease – molecular scissor → correct

C. DNA polymerase – joins nucleotides (not nucleosides) → incorrect

D. RNA polymerase – joins nucleotides → correct

Final answer: Option 3

112) **Question Explanation:** Identify the incorrectly matched pair related to genetic engineering methods and tools.
Concept: Gene transfer methods/tools.
Solution:

A. Microinjection → animal cells - Correct
B. Biolistic method → plant cells - Correct
C. Plasmid → Virus Incorrect (plasmids are circular DNA molecules, not related to viruses)
D. EcoRI →is a restriction endonuclease enzyme - Correct

Final answer: (3)

113) **Question Explanation:** The question asks to select the correct matches between molecular biology terms and their descriptions.
Concept: Restriction enzymes, cDNA, PCR, gene transfer, and diagnostic methods
Solution:
(a) First restriction endonuclease → Hind II – True, Hind II was the first discovered restriction enzyme.
(b) EcoR I and EcoR V producing sticky ends – False; Eco RI produces sticky ends, but Eco RV produces blunt ends.
(c) cDNA → Shorter than original gene – True, as introns are removed during reverse

transcription.
(d) Direct method of gene transfer → *Agrobacterium* mediated – False, Agrobacterium is indirect method.
(e) Conventional methods of diagnosis → Serum and urine analysis – True.
(f) PCR → RNA primers – False; PCR uses DNA primers, not RNA primers.
Final Answer : (3)

114) **Explain Question –** The question asks which option correctly fills all blanks related to restriction enzymes, cell wall–digesting enzymes, recombinant protein, and DNA staining.
Concept – Complete biotechnology : Principles and processes
Solution – Same restriction enzyme → same sticky ends
Joining DNA fragments → DNA ligase
Cell wall digestion enzymes → lysozyme (bacteria), cellulase (plants), chitinase (fungi)
Protein expressed in a different host → recombinant protein
DNA staining → ethidium bromide
Final answer : (4)

Option 4

115) **Solution :** a and c are correct
(a) Inactivation of amp^R gene.
(c) Unable to grow in presence of ampicillin.
Final Answer : (2)

116) **Explain Question :** We must identify which two statements among A–D are incorrect.
Concept –Transgenic Animals
Solution – (A) Wrong → Rosie is a cow, not a buffalo.
(B) Wrong → Restriction enzymes do not isolate DNA.
Final answer : (1)

117)

Question Explanation: Identify what colonies produce when plasmid has no insert (blue-white screening).
Concept: Biotechnology
Solution:

A. Option 1: Correct → If plasmid has no insert, lacZ gene remains functional and produces β-galactosidase, which breaks X-gal and forms blue colonies.
B. Option 2: Incorrect → No colour appears when lacZ is disrupted (insert present).
C. Option 3: Incorrect → X-gal is required for blue colour formation.
D. Option 4: Incorrect → Maltose has no role in this test.

Final Answer: Option 1

118) **Solution :** Lysozyme, Chitinase and Cellulase are used for isolation of DNA.
Final Answer : (3)

119)

Question Explanation: Asks the technique where foreign DNA is directly injected into the nucleus of an animal cell.
Concept: Gene transfer method (Microinjection)
Solution: The correct answer is option 2

A. Micro-injection uses a fine needle to directly inject DNA into the nucleus of animal cells or embryos.
Other methods like biolistics and heat-shock are used mainly for plant or bacterial cells.

Final Answer: Option 2

120) **Question Explanation:** The question is asking which one of the given pairs correctly matches the term with its description.
Concept: Insertional inactivation and identification of recombinants
Solution: In insertional inactivation, when a gene is inserted into a selectable marker site, that marker loses its function. This helps differentiate recombinants from non-recombinants, which matches the given description correctly. Recombinant proteins are produced in heterologous host.
Downstreaming is the process done after completion of biosynthetic stage. Cloning is making multiple copies of a gene.
Final Answer : (1)

121) **Explain Question:** Which option correctly identifies components of pBR322?
Concept: Cloning Vectors
Solution : BamH-I, EcoRI - Recognition sites for restriction enzymes
Ori → Site from where replication starts, amp^R, tet^R - Selectable marker.
rop gene → codes for a protein involved in regulation of plasmid replication.Other options are incorrectly defined.
Final answer : (3)

122) **Solution :** In this cells are maintained in their physiologically most active log phase of growth.
Final Answer : (2)

123) **Question explanation:** Asks to arrange the steps of recombinant DNA technology.
Concept: Steps of recombinant DNA technology
Solution: The correct order of steps of RDT are –
(III) Isolation of DNA fragment
(II) Amplification of gene of interest.
(IV) Ligation of DNA fragments into vector.
(V) Insertion of r-DNA into host.
(I) Extraction of desired gene product.
Final Answer : (3)

124)

Question Explanation: Asks which components are present in a typical bioreactor.
Concept: Bioreactor – essential components
Solution : A standard bioreactor includes:

A. Agitator system (mixing)

B. Oxygen delivery system (aeration)

C. Foam control system

D. Temperature control system

E. pH control system

F. Sampling ports (monitoring)

All listed components are correct.
Final Answer : (4)

125) **Solution :** Smaller the DNA fragment, farthest it moves.
Final Answer : (4)

126) **Solution :** A-Denaturation, B - Annealing, C - Extension.
Final Answer : (1)

127) **Solution :** In a continuous culture system, the cells are maintained in their physiologically most active log phase.
Final Answer : (3)

128) **Solution :** DNA can be visualized after staining with ethidium bromide followed by exposure to UV light.
Final Answer : (4)

129) **Question Explanation:** Asking about agarose gel electrophoresis
Concept: Agarose gel electrophoresis
Solution: Both assertion and Reason are correct but reason is not correct explanation of Assertion
Final Answer : Option (2)

130) **Question Explanation –** The question is asking whether the given Assertion and Reason

are true, and whether the Reason correctly explains the Assertion.
Concept – Mode of action of Bt toxin and specificity of Bt proteins
Solution – The Assertion is true: the activated Bt toxin binds to receptors on the midgut epithelial cells of insects, forms pores, causes cell swelling, lysis, and leads to insect death.
The Reason is also true: most Bt toxins are insect-group specific because each toxin binds only to specific receptors present in certain insect groups.
However, the Reason does not explain why the toxin forms pores and causes cell lysis. It explains specificity, not the mechanism described in the Assertion.
Final Answer : (1)

131) **Solution:** Both statements I and statement II are correct.
Final Answer : (3)

132) **Solution:** Statement I : The size of a population for any species is not a static parameter
Final Answer : (2)

133) **Solution:** Both Statement I and Statement II are correct, as both the statements explains natality and mortality.
Final Answer : (3)

134) **Solution:** Statement II : The density of a population in a given habitat during a given period fluctuates due to changes in four basic processes, i.e, natality, mortality, immigration and emigration.
Answer: (1)

135) **Solution:** Statement I: If N is the population density at time t, then its density at time t +1 is $N_{t+1} = N_t + [(B + I) - (D + E)]$.
Final Answer : (2)

136) **Solution:** Biological control uses natural predators to check pests.
Final Answer : (2)

137) **Solution:** Population = group of interbreeding individuals of a species; community = all populations in a given area.
Final Answer : (1)

138) **Solution:** Tiger and deer show predation, not mutualism.
Final Answer : (4)

139) **Solution:** Hormonal secretion is a property of an individual, not a population.
Final Answer : (4)

140) **Solution:** Defences reduce palatability or cause harm, not increase nutrition.

Final Answer : (3)

141) **Solution:** Population stabilises when resources are limited.
Final Answer : (2)

142) **Solution:** *Balanus* competitively excludes *Chthamalus* from lower shore.
Final Answer : (1)

143) **Solution:** Species minimize competition by using different resources.
Final Answer : (2)

144) **Solution:** Predators regulate prey populations, preventing dominance of one species.
Final Answer : (1)

145) **Solution:** Desert plants reduce leaf area or have modified stems.
Final Answer : (3)

146) **Solution:** Primary productivity depends on plant species and environmental factors like temperature and sunlight. It is expressed in kcal m^{-2} yr^{-1}, not dependent on it.
Final Answer : (2)

147) **Solution:** Resistance and resilience are properties of stable ecosystems and are positively influenced by species diversity.
Final Answer : (1)

148) **Solution:** Detritivores fragment detritus; decomposers convert complex organic matter; humification forms humus; mineralisation releases inorganic nutrients.
Final Answer : (1)

149) **Solution:** Each pair represents an interaction type as described in NCERT examples.
Final Answer : (1)

150) **Solution:** GPP is total energy fixed in photosynthesis before respiration losses.
Final Answer : (1)

151) **Solution:** Standing crop refers to biomass, while standing state refers to nutrient pool.
Final Answer : (1)

152) **Solution:** Under unlimited resources, exponential growth occurs.
Final Answer : (2)

153) **Solution:** Tigers regulate herbivore populations, maintaining balance in forests.

Final Answer : (1)

154) **Solution:** Higher species richness leads to greater stability and resilience.
Final Answer : (2)

155) **Solution:** Decomposers convert organic matter into inorganic forms usable by producers.
Final Answer : (2)

156) **Solution:** Keystone species have high functional impact despite low abundance.
Final Answer : (3)

157) **Solution:** Litter rich in nitrogen and poor in lignin decomposes faster.
Final Answer : (3)

158) **Solution:** Bacteria rapidly grow due to nutrient availability but later stabilise as resources deplete.
Final Answer : (1)

159) **Solution:** A–i, B–iii, C–ii, D–iv

A. Fragmentation	i. Breakdown of detritus into small particles by detritivores
B. Leaching	iii. Water-soluble nutrients lost from detritus
C. Mineralisation	ii. Conversion of complex organic matter into inorganic forms
D. Humification	iv. Formation of dark, amorphous, stable organic matter

Final Answer : (1)

160) **Solution:** Diverse litter provides multiple substrates for different decomposer microbes.
Final Answer : (1)

161) **Solution:** Pelagic fishes and oyster produce numerous small offspring; birds and mammals produce fewer large offspring.
Final Answer : (3)

162) **Solution:** Logistic growth shows lag, exponential, deceleration, and asymptote at carrying capacity.
Final Answer : (3)

163) **Solution:** Rainfall is true but biodiversity importance is the main reason.
Final Answer : (2)

164) **Solution:** Both true, but unrelated explanation.

Final Answer : (2)

165) **Solution:** A and E incorrect; B, C, D correct.
Final Answer : (2)

166) **Solution:** All are NCERT facts.
Final Answer : (4)

167) **Solution:** Variation drives evolution → biodiversity.
Final Answer : (1)

168) **Solution:** Cross breeding is not conservation.
Final Answer : (3)

169) **Solution:** Direct NCERT examples.
Final Answer : (1)

170) **Solution:** Cryopreservation preserves genetic material.
Final Answer : (3)

171) **Question Explanation:** This question asks us to identify plant groups A, B, and C in a diagram showing the proportion of species among major taxa.

Concept: Number Of Species On Earth And Pie Charts - Invertebrate, Vertebrate, Plants.

Solution:
A (smallest) → Algae (fewer species),
B (moderate) → Lichens,
C (largest) → Angiosperms (flowering plants, most diverse)

Final answer: 4

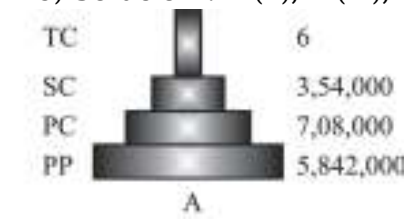
172) **Question Explanation :** The question asked to identify the correct graph representing the species-area relationship **Concept :** This question is based on Patterns of Biodiversity.
Solution : Only option 3 shows a correct species - area relationship : logC + ZlogA
• X - axis : Area
• Y - Axis : Species richness
• Shape : steep rise followed by gradual leveling.
Final Answer: (3)

173) **Solution :** Other animal groups → Crustaceans → Molluscs → Insects.
Final Answer : (2)

174) **Question Explanation:** Identify the principle that the upright ecological pyramid follows

without exception.
Concept: Pyramid of Energy
Solution: The given pyramid is an upright, tapering pyramid, representing the decrease of a parameter across trophic levels.
This shape is universally true for the pyramid of energy, as energy is lost at each step (Second Law of Thermodynamics), preventing inversion.
Final Answer : (2)

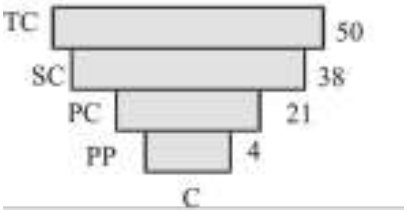
175) **Solution :** A-(ii), B-(iii), C-(i)



(ii) Pyramid of numbers in a grassland ecosystem showing about 6 million producers.



(iii) Upright pyramid of biomass.



(i) Inverted pyramid of biomass depicting small standing crop of phytoplanktons supporting a large standing crop of zooplanktons.
Final Answer : (1)

176) **Solution :** The given diagram shows, relationship between fig. and wasp (Mutualism).
Final Answer : (1)

177)

Question Explanation: The question is asking: "Look at the given figure and identify which plant (from the options) is pollinated by an insect."



Concept: Mutualism

Solution: Orchid flower is the correct answer because orchids have special, complex flowers that attract insects for pollination. They often use bright colors, unique shapes, and scents to bring insects, and many orchids even use sexual deceit to trick insects into pollinating them.So the figure matches an insect-pollinated orchid flower.

Final answer: The correct answer is option 1. Orchid flower

178) **Question Explanation:** Relationship between diversity and stability.
Concept: Diversity-stability hypothesis.
Solution: According to David Tilman's research, a species-rich community is more stable because it shows less year-to-year variation in biomass, has higher productivity, and is more resistant to invasions by alien species. All three options are correct indicators of stability in a diverse ecosystem.
Final Answer : (1)

179) **Question Explanation:** Matching the rationales for biodiversity conservation with their correct categories.
Concept: Reasons for Biodiversity Conservation
Solution: The reasons for conserving biodiversity are grouped into three categories:

A. Narrowly utilitarian: Direct economic benefits like food, firewood, and medicines. → (ii) Direct economic benefits.
B. Broadly utilitarian: Indirect, major ecosystem services like oxygen production, pollination, and aesthetic pleasure. → (i) Ecosystem services (oxygen, aesthetic value).
C. Ethical: Based on moral duty to protect every species, acknowledging their intrinsic value. → (iii) Intrinsic value (moral duty).

Final Answer : (2)

180) **Question Explanation :**The question asks you to analyze two statements — an Assertion and a Reason — and then choose the correct option that explains their truthfulness and relationship.
Concept : - Loss Of Biodiversity
Explanation :
Assertion: True. Amazon is widely referred to as "lungs of the planet."
Reason: True. The Evil Quartet is indeed related to biodiversity conservation.
The reason (Evil Quartet) is not the explanation for why the Amazon is called lungs of the planet.
Final answer : (2)